

How Knowledge Crosses Borders: Evidence from Semiconductor Manufacturing^{*}

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Abstract

We show that cross-border learning is an important feature of modern semiconductor manufacturing and that several important channels – such as knowledge sharing along global supply chains, the movement of engineers within multinational firms, and foreign direct investment – operate through market mechanisms rather than anonymous technological spillovers. These findings have significant policy implications. When cross-border learning is mediated by purposeful economic relationships, access to frontier knowledge is neither automatic nor guaranteed. Firms and governments can facilitate - or restrict - the international diffusion of technology, implying that policies to foster domestic technological upgrading are likely to be most effective when coupled with continued access to global sources of frontier knowledge.

JEL: F50, F61, L63, N60, O30

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1 Introduction

Semiconductor manufacturing has become the focus of unprecedented government intervention around the world. Motivated by concerns about resilience, national security, and technological leadership, governments have committed substantial resources to expanding domestic production while increasingly restricting the international diffusion of advanced technologies. These policies rest on the widely accepted view that semiconductor manufacturing exhibits strong learning-by-doing: countries that establish an early lead acquire technological advantages that are difficult for others to replicate, making government support for domestic production an attractive policy instrument.

We argue that this perspective is incomplete because it implicitly treats learning-by-doing as a process taking place within national borders. While experience accumulated through production within a country is undoubtedly important, technological progress also depends on access to knowledge generated abroad. Modern semiconductor manufacturing is embedded in a highly international production network, in which firms, workers, suppliers, and production facilities span national borders. As a result, technological progress depends critically on cross-border learning. Understanding how learning occurs is therefore essential for evaluating the effectiveness of current policy initiatives. This observation has received relatively little attention in policy discussions, which have focused primarily on the location of production rather than the location and transmission of knowledge.

In this paper, we show that cross-border learning is important and provide evidence that several of its key channels operate through market mechanisms rather than through geographic proximity or other passive forms of knowledge diffusion. Knowledge moves across borders through interactions along global supply chains, international labor mobility, multinational firms, and foreign direct investment. These channels reflect deliberate organizational and business decisions rather than incidental contact among nearby firms. As a consequence, international knowledge transmission is neither inevitable nor irreversible: it can be facilitated or disrupted by the decisions of firms and governments alike. To the extent that firms capture some of the benefits from transmitting knowledge when organizing production, investment, and hiring across borders, they may internalize part of the private returns to these activities. Governments, however, pursue objectives defined at the national level and therefore need not internalize the benefits that international knowledge transmission creates outside their own borders. This divergence creates a potential tension between the incentives of globally integrated firms and the objectives of national governments.

We establish this result using three complementary approaches. First, we review industry evidence highlighting the importance of access to foreign knowledge in semiconductor manufacturing. The industry's historical record reinforces this view as it reveals that with the exception of the United States, where the industry originated, every major semiconductor

producing country built its manufacturing capabilities through sustained access to frontier technologies from abroad. Second, we develop and estimate a structural model of semiconductor manufacturing that quantifies the importance of cross-border learning. Third, we present reduced-form evidence on some of the principal channels through which knowledge is transmitted internationally, focusing on learning along the global supply chain, labor mobility, including within multinational firms, and foreign direct investment. Although these three approaches rely on different data and methodologies, they all point to the same conclusion: international knowledge transmission is a fundamental feature of modern semiconductor manufacturing and important channels of cross-border transmission are market-mediated.

Our findings have important implications for current policy debates. If knowledge is transmitted through international market relationships, policies that strengthen or weaken those relationships can have powerful effects on technological development. On the one hand, policies aimed at supporting the domestic industry are likely to be most effective when coupled with continued access to global sources of frontier knowledge. On the other hand, export controls and other measures designed to slow technological catch-up by geopolitical rivals can be highly effective precisely because they disrupt the market mechanisms through which frontier knowledge is transmitted. Our results also point to a potential tension between firms and governments. Firms have incentives to maintain international production networks when they internalize the benefits of knowledge sharing across borders. Governments pursuing resilience or national security objectives may instead seek to reduce dependence on foreign production or foreign technology, even when doing so disrupts the channels through which knowledge is transmitted. We do not take a position on whether such objectives are desirable; rather, our analysis highlights the tradeoffs that arise once international knowledge transmission is recognized as an important feature of the industry.

More broadly, our results suggest a different way of thinking about knowledge diffusion in globally integrated industries. Rather than viewing technological progress as arising primarily from localized interactions, we emphasize the role of firms and international production networks in actively and deliberately transmitting knowledge across countries. This perspective has implications that extend beyond the semiconductor industry and highlights the importance of global production networks in shaping the geography of technological progress.

Related literature

Our paper contributes to several strands of literature. First, it relates to the literature documenting learning-by-doing in technologically intensive industries and its role in shaping industry dynamics (e.g., [Arrow \(1962\)](#); [Irwin and Klenow \(1994\)](#); [Benkard \(2000\)](#); [Besanko, Doraszelski, Kryukov, and Satterthwaite \(2010\)](#); [Thurk \(2024\)](#)). Our contribution is to show that an important component of this learning is transmitted across national bor-

ders through firms' international production networks rather than generated solely through domestic production.

Second, the paper contributes to the broader literature on knowledge diffusion (e.g., Bloom, Schankerman, and Van Reenen (2013)) and international technology transfer (e.g., Coe and Helpman (1995); Keller (2002, 2004)). Previous work has emphasized a variety of channels through which knowledge crosses borders, including multinational firms (e.g., Markusen (2002); Javorcik (2004); Alfaro and Chen (2018)), foreign direct investment (e.g., Aitken and Harrison (1999)), and the international mobility of skilled workers (e.g., Almeida and Kogut (1999); Oettl and Agrawal (2008); Poole (2013)). We build on these insights by showing that these channels operate jointly in semiconductor manufacturing and are embedded in firms' international production networks. We further argue that global supply chains play an important role in transmitting technological know-how across countries. While the literature on global value chains has primarily emphasized the fragmentation and organization of production (e.g., Antràs and Chor (2013); Antràs (2021)), we emphasize that these same production networks also function as mechanisms for cross-border learning, a point also made in the World Bank (2020) *World Development Report* on global value chains. More broadly, our paper bridges two largely separate literatures—international technology diffusion and global value chains—by showing that production networks are not only organizational arrangements for coordinating production, but also mechanisms for transmitting technological capabilities across countries.

Third, the paper contributes to the growing literature on semiconductor policy and industrial policy more broadly (e.g., Bown (2020); Bonvillian and Sarma (2023); Bonvillian (2024); Miller (2022); Juhász, Lane, and Rodrik (2024); Goldberg, Juhász, Lane, Lo Forte, and Thurk (2024); Liu, Rotenberg, and Traiberman (2026); OECD (2019)). Much of this work has focused on the consequences of subsidies, export controls, and efforts to strengthen domestic production. A recent literature has shown that industrial and trade policies can have unintended consequences through their effects on innovation and technology adoption. Alfaro, Fadinger, Schymik, and Virananda (2025), for example, show that China's restrictions on rare-earth exports induced foreign firms to innovate in ways that reduced their dependence on rare earths, thereby mitigating the intended effects of the policy. Our analysis contributes to this literature by emphasizing that policy effectiveness depends critically on how knowledge is transmitted across countries. Policies that alter firms' ability to participate in international production networks affect not only the location of production, but also the diffusion of technological capabilities.

Finally, the paper relates to the emerging literature on geoeconomics (e.g., Farrell and Newman (2019a); Clayton, Maggiori, and Schreger (2026)), which studies the interaction between global production networks and governments' pursuit of national strategic objectives. This literature emphasizes that firms and governments may face different incentives when

international economic relationships create both economic gains and strategic vulnerabilities. Our contribution is to identify international knowledge transmission as one important channel through which the incentives of globally integrated firms and nationally oriented governments may diverge. When firms internalize the private benefits of organizing production and knowledge flows across borders while governments pursue objectives defined at the national level, international knowledge transmission may generate incentives for firms and governments that are not fully aligned.

The remainder of the paper is structured as follows. Section 2 provides a brief overview of the industry focusing on the role of learning and knowledge transmission. Section 3 presents the empirical model used to estimate learning, introduces the proprietary data used to estimate the model, presents the empirical findings, and explores their policy implications through two counterfactual exercises. Section 4 explores the mechanisms behind cross-border learning, and section 5 concludes.

2 Learning and Knowledge Transmission in Semiconductor Manufacturing

The central technological feature of semiconductor manufacturing is that production itself generates knowledge. Fabricating chips involves a long sequence of highly precise operations whose performance cannot be determined entirely from blueprints, scientific principles, or the equipment used in production. Manufacturers improve yields and reduce costs by repeatedly operating production lines, diagnosing defects, adjusting process parameters, and incorporating what they learn into subsequent production. In this sense, semiconductor manufacturing is a canonical setting for learning-by-doing (Weber, 2004; Guldi, 2004; Irwin and Klenow, 1994).

At the same time, the organization of the modern semiconductor industry makes the relevant unit of learning less obvious than the individual production line or even the individual firm. Semiconductor production has become highly specialized and internationally fragmented (Kapoor and McGrath, 2014; Varas, Varadarajan, Goodrich, and Yinug, 2021; Bown, 2020). Foundries rely on equipment manufacturers, materials suppliers, chip designers, and electronic design automation firms, many located in other countries. Moreover, the boundaries between these activities are technologically porous: improving one part of the production process frequently requires knowledge possessed by firms operating at another stage of the value chain (Appleyard, 2003). As a result, a foundry's productive capabilities may reflect not only its own accumulated manufacturing experience, but also experience acquired with other technologies within the firm and knowledge generated elsewhere in the global industry.

These observations motivate the empirical framework developed in the next section. We distinguish three sources of manufacturing experience: experience accumulated by a foundry producing a particular technology; experience accumulated by the same foundry producing other technologies; and experience accumulated by other producers of the same technology around the world. The discussion in this section provides the technological and historical basis for this decomposition. Section 3 estimates the contribution of each source of experience to production costs, while Section 4 examines some key mechanisms through which knowledge generated abroad reaches domestic producers.

2.1 Process knowledge and learning from production

Semiconductor fabrication begins with the construction of networks of transistors on a silicon wafer. Circuit patterns are transferred onto the wafer using photolithography and built up layer by layer through repeated operations including deposition, etching, doping, and cleaning. After fabrication, the wafer is tested to determine which individual chips, or dies, function correctly; it is subsequently cut into individual dies, which are packaged and integrated with other components. Technological progress has historically been associated with the ability to manufacture increasingly small and dense transistors, producing the sustained improvements in computing performance associated with Moore's law; achieving each successive generation requires advances across multiple scientific and engineering domains (Varas et al., 2021; Pillai, 2023, pp. 8, 52).

What matters for learning-by-doing is the extraordinary complexity and precision of this manufacturing process. A finished wafer may pass through hundreds—and for advanced processes more than one thousand—individual production steps over several months; a defect introduced at an early stage can waste the work embodied in every subsequent step (Varas et al., 2021, pp. 16, 21). Individual pieces of equipment themselves require the simultaneous adjustment of many operating parameters, while those parameters must in turn be coordinated with the settings of other machines throughout the production line. The equipment embodies frontier scientific knowledge (Pillai, 2023, p. 965), but is necessary rather than sufficient for successful production: even a single etch module can involve more than fifty settings that must be tuned jointly with the rest of the process (Appleyard, 2003, pp. 359–360).

What manufacturers accumulate through production is consequently a form of *process knowledge*: knowledge about recipes, machine settings, sequencing, defect detection, and the interactions among different stages of fabrication. An important manifestation of this knowledge is the *yield*—the fraction of dies produced on a wafer that function as intended. Engineers in the industry refer to the systematic improvement in this outcome as *yield learning*. Weber (2004) describes yield learning as a procedure that every manufacturer must master and whose pace is central to profitability. Improving yields requires identifying sources of defects,

experimenting with alternative settings, and incorporating the results of those experiments into subsequent production; sustained defect reduction has become a fundamental element of wafer-fab practice (Guldi, 2004). The accumulated knowledge is partly tacit and difficult to codify; consistent with this interpretation, semiconductor manufacturers have historically relied heavily on secrecy rather than patents to protect process innovations (Cohen, Nelson, and Walsh, 2000).

This is the first and most direct form of learning represented in our empirical framework. Repeated production of a particular technology k by a particular foundry j generates cumulative experience for this foundry-technology pair Y_{jkt} at time t . Higher cumulative output of this specific technology provides more opportunities to identify defects and refine the manufacturing process and therefore lowers subsequent production costs. This is the conventional learning-by-doing mechanism emphasized in earlier work on semiconductors, including Irwin and Klenow (1994).

But process knowledge need not be specific to a single technology. Many capabilities acquired in operating a fab—diagnostic routines, process-control methods, organizational procedures, and knowledge about common pieces of equipment—can be useful when the firm begins producing a different generation of chips. The simultaneous operation of multiple technology nodes by large foundries illustrates the scope for this type of transfer. TSMC, for example, routinely begins ramping production of a new node while continuing to manufacture several preceding generations; the evolution of its sales across nodes shown in Figure 1 illustrates the overlapping nature of these technology cycles. The model allows for this second margin of learning through cumulative experience acquired by the same foundry across all technologies it has produced in the past.

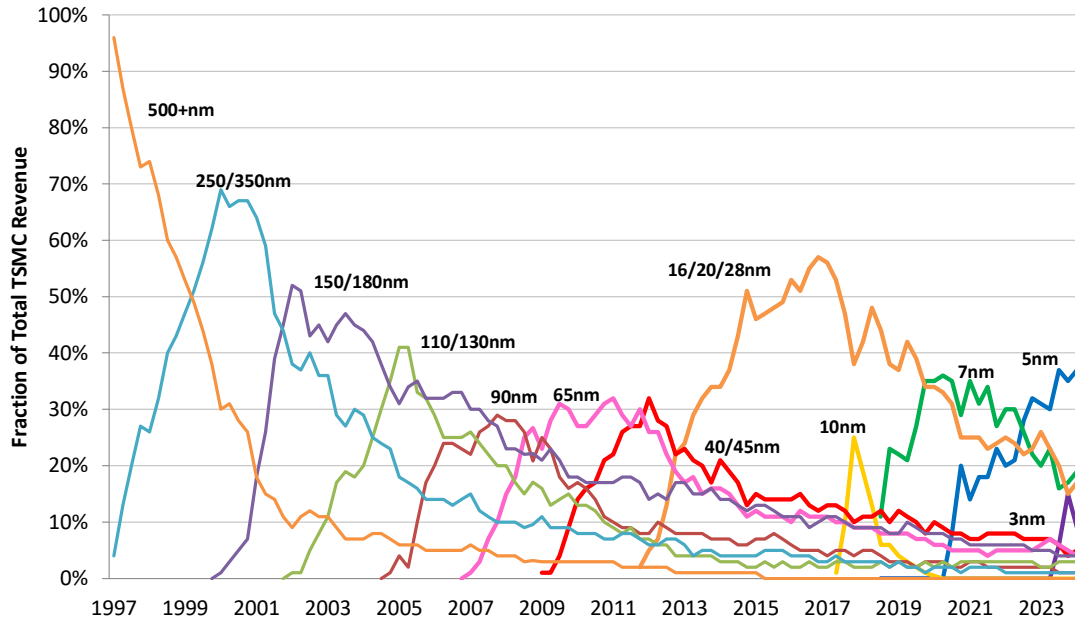
2.2 Specialization and the limits of within-firm learning

If all technologically relevant knowledge were generated and retained within a vertically integrated manufacturer, these two forms of internal learning might provide an adequate description of semiconductor production. The organization of the industry, however, has evolved in precisely the opposite direction.

Early semiconductor producers such as AT&T, Fairchild, IBM, and Texas Instruments designed much of their own manufacturing equipment. Specialized equipment producers began to emerge already in the late 1950s, often with technology and know-how deliberately transferred to them by semiconductor manufacturers (Pillai, 2023, pp. 960, 976). As the scientific complexity of production increased and the investment required to remain at the frontier grew, specialization spread throughout the value chain (Varas et al., 2021, p. 23).

Today, the industry consists of several highly specialized groups of firms (Varas et al., 2021; Bown, 2020). Fabless firms such as NVIDIA and Qualcomm design chips but do not

Figure 1: TSMC Sales by Line Width

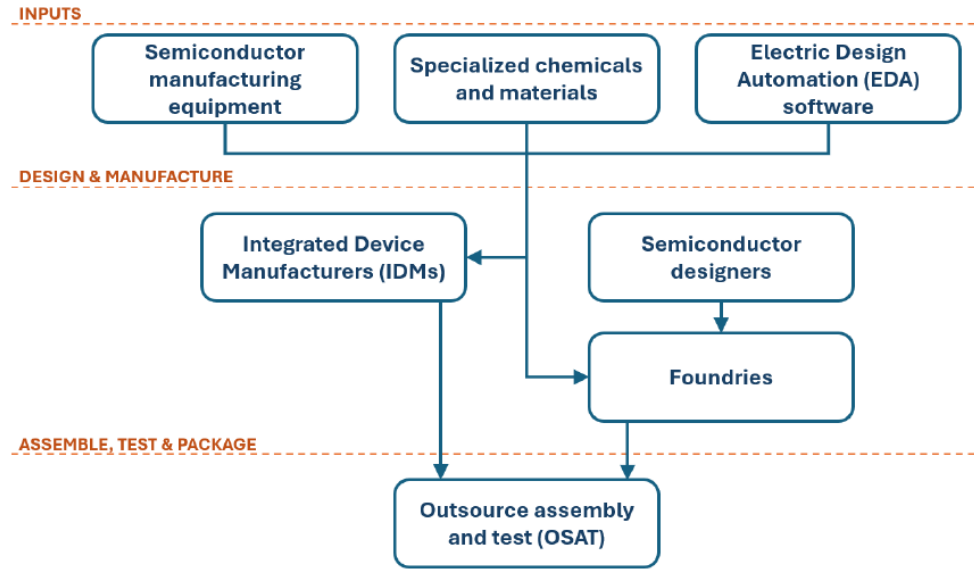


manufacture them. Foundries such as TSMC manufacture chips designed by other firms. Integrated device manufacturers such as Intel and Samsung combine design and manufacturing, while outsourced assembly and test firms perform downstream packaging and testing. These producers in turn rely on specialized EDA software companies, manufacturing-equipment producers, and suppliers of wafers, chemicals, and other materials. The resulting supply chain, illustrated in Figure 2, is both international and highly geographically concentrated (Varas et al., 2021; OECD, 2019; Bown, 2020).

Specialization does not, however, imply technologically independent stages of production. Quite the opposite: as firms have become narrower in what they produce, successful innovation has increasingly required coordination across organizational boundaries. A piece of advanced manufacturing equipment cannot simply be delivered to a fab and operated independently of the rest of the production process. Its performance depends on matching the machine's capabilities to the fab's process conditions, while neither party possesses all the information required to perform this optimization alone (Appleyard, 2003). This kind of interdependence is a general feature of complex technologies, in which progress in one component can expose bottlenecks in another (Rosenberg, 1963; Hughes, 1983).

The interaction between lithography-equipment producers and fabs provides a useful illustration. IBM and ASML engineers, for example, jointly developed methods for controlling critical dimensions in IBM's production facility, combining information obtained from wafers in the fab with correction capabilities embedded in ASML's equipment (Wong, Seevaratnam, Wiltshire, Felix, Brunner, Rawat, Escalante, Elmalk, Wang, Leewis, Hinnen, Kim, and Rottenkolber, 2013). Similar collaboration between ASML and SK Hynix combined

Figure 2: The Semiconductor Supply Chain:
Highly Fragmented, Global, and Geographically Concentrated



Simplified version of Bown (2020)

measurements generated on the manufacturer's production line with information and software controlled by the equipment producer (Kim, Ma, Kim, Kim, Kwon, Park, Nikolsky, Otter, Escalante Marun, Anunciado, Sun, Storms, and van West, 2014). These examples are not instances in which knowledge simply diffuses because two firms happen to be geographically close. They involve deliberate cooperation between firms possessing complementary pieces of technological knowledge.

The same principle applies elsewhere in the value chain. EDA firms translate information about what manufacturing processes can reliably produce into design rules that chip designers can use (Liebmann, Kye, Kim, Yuan, and Geronimi, 2010; Ma, Sweis, Yoshida, Wang, Kye, and Levinson, 2012). Equipment suppliers adapt machines to evolving requirements at the technological frontier, while foundries work closely with customers and suppliers to reconcile product design with manufacturability (Appleyard, 2003). The relevant technological knowledge is therefore distributed across firms: no single participant necessarily possesses a complete understanding of the production system.

This interdependence becomes particularly important at the technological frontier. The development of extreme-ultraviolet lithography, for example, required decades of research involving universities, public laboratories, ASML, and several of the world's leading semiconductor manufacturers; the technology took almost four decades to move from early concepts to commercial implementation (Varas et al., 2021, p. 14). Intel, Samsung, and

TSMC—competitors in semiconductor manufacturing—participated alongside ASML in its development and commercialization (Varas et al., 2021, p. 29). A modern EUV machine itself incorporates roughly 100,000 parts supplied by more than 5,000 firms (Varas et al., 2021, pp. 29–30). More generally, collaborative R&D in semiconductor manufacturing tends to concentrate on technological bottlenecks (Kapoor and McGrath, 2014, pp. 557–558).

Specialization can therefore create a powerful mechanism for transmitting knowledge. When a supplier develops an improvement together with one customer and subsequently incorporates that improvement into equipment, software, or operating practices supplied to other customers, knowledge generated in one production relationship becomes useful elsewhere. Equipment suppliers can in this sense operate as “conduits of knowledge” between manufacturers (Appleyard, 2003, p. 359). The knowledge may cross both firm boundaries and national borders.

This observation changes the interpretation of learning-by-doing in a globally fragmented industry. Production by one manufacturer may contribute to improvements that eventually affect the productivity of another manufacturer—not necessarily because the second firm observes or imitates the first, but because both participate in a production network containing common suppliers, customers, workers, and other intermediaries.

2.3 Cross-border learning

The international organization of semiconductor production makes this distinction especially important. Advanced design and EDA capabilities remain concentrated in the United States; the most sophisticated manufacturing equipment is produced by a small number of firms in the Netherlands, Japan, and the United States; and much of semiconductor fabrication takes place in East Asia (Varas et al., 2021; OECD, 2019; Bown, 2020). Modern chip production therefore requires repeated economic interactions among firms located in different countries.

Process knowledge can cross national borders through several such relationships. Multi-national semiconductor firms can deliberately transfer knowledge among their own establishments, including by moving experienced engineers across plants. Workers can also carry tacit knowledge when they change employers; in semiconductors, patenting patterns have been shown to follow engineers as they move across firms (Almeida and Kogut, 1999). Equipment and EDA suppliers interact with multiple manufacturers in different countries. Fabless customers work with foundries to adapt designs to manufacturing processes and may subsequently work with other foundries. Foreign direct investment provides another organizational structure through which technological and managerial capabilities developed in one location can be deployed in another.

The distinction between these mechanisms and an anonymous technological spillover is important for the interpretation of our analysis. We do *not* require the knowledge accumulated

through production in one country to become freely available to every producer elsewhere. Rather, accumulated global production creates a stock of potential knowledge, access to which may depend on participation in particular economic relationships. The empirical model we develop in the next section deliberately abstracts from these individual transmission mechanisms. It summarizes their combined effect through the cumulative experience of other producers of the same technology, and estimates a parameter governing how strongly that external experience affects the costs of a specific foundry.

Accordingly, the cross-border term in the model should be interpreted as *reduced-form cross-border learning*, rather than as a literal assumption that knowledge is “in the air.” Section 4 will investigate the mechanisms underlying this term. This interpretation is central to the argument of the paper: international experience can matter greatly for productivity even when access to the knowledge generated by that experience is mediated by firms, supplier-customer relationships, workers, and investment.

Taken together, the technological structure of the industry therefore suggests three distinct margins of learning that correspond directly to the experience stock used in the model: The first captures conventional learning-by-doing. The second captures economies of scope in learning within a foundry. The third captures knowledge generated through production elsewhere that becomes accessible through international production and knowledge networks. Section 3 estimates the relative importance of these three sources of experience.

2.4 From the industry evidence to the empirical model

The discussion in this section suggests a broader interpretation of the conventional semiconductor “learning curve.” Manufacturing experience clearly matters, but the relevant experience need not coincide with cumulative production of one product by one producer. Process improvements can carry across technologies within a foundry, while the specialization and international fragmentation of semiconductor production allow knowledge generated through production elsewhere to enter a firm’s technological opportunity set.

This motivates the experience measure used in Section 3, which allows costs to depend separately on own-technology experience, experience accumulated in other technologies within the same foundry, and experience accumulated by other producers of the same technology around the world. The model is deliberately agnostic about the precise mechanism underlying the last component. The “cross-border-learning” parameter we estimate measures whether, and to what extent, production experience accumulated elsewhere predicts lower costs for a given producer after controlling for its own experience and experience elsewhere within the firm. Section 4 then asks *how* such international knowledge transmission takes place.

This distinction between *the existence of cross-border learning* and *the mechanisms that give firms access to it* is essential. The model quantifies the former. The evidence on supplier-customer relationships along the global supply chain, multinational production, worker mobility, and foreign direct investment will be used to investigate the latter. In combination, the two pieces of the analysis allow us to distinguish a world in which firms become more productive simply because relevant knowledge diffuses passively from one location to another from one in which international production networks themselves constitute an important part of the technology through which knowledge is transmitted.

3 Model-Based Analysis

3.1 A Simple Industry Model

This section describes the model, the estimation procedure, identification assumptions, and empirical results, and discusses their implications for within-firm versus cross-firm learning. The theory combines several areas of economic literature. First, we follow the industrial organization literature in the modeling of international competition in strategic industries (e.g., Kalouptsi (2018); Barwick, Kalouptsi, and Zahur (2025); Miravete, Moral, and Thürk (2018)). Second, our demand system leverages the international trade literature’s approach to estimating tractable demand models that account for cross-country quality differences (e.g., Hummels and Klenow (2005); Khandelwal (2010); Hallak and Schott (2011); Kugler and Verhoogen (2011); Johnson (2012); Fajgelbaum, Goldberg, Kennedy, and Khandelwal (2020); Fajgelbaum, Goldberg, Kennedy, Khandelwal, and Taglioni (2024)). Finally, we incorporate dynamics by leveraging the empirical learning-by-doing literature (e.g., Irwin and Klenow (1994); Benkard (2000); Besanko, Doraszelski, and Kryukov (2019)).

The model focuses on the relationship between foundries where chips are produced—the suppliers in the model (e.g., TSMC in Taiwan or SMIC in China)—and the fabless firms who purchase chips—the buyers in the model (e.g., NVIDIA)—from foundries. Each period buyers (e.g., NVIDIA) source their chips from countries around the world. We assume that each country is endowed with a single firm which produces chips (e.g., TSMC in Taiwan and SMIC in China).¹ All period t information is known to buyers and firms but firms form

1. In practice, this assumption is less restrictive as it may seem, due to the high concentration in this industry – for instance, in Taiwan, TSMC has been the dominant player; in China, SMIC is the largest foundry by a wide margin.

expectations given period t information. Dynamics matter only through the supply-side via learning by doing. We further assume that buyers are myopic.²

3.2 Demand

We assume that a representative buyer (customer) maximizes a 3-tier nested CES objective function over technologies k and suppliers j in each quarter t . “Technology” k here refers to wafer diameter and the line width of the IC chips. More advanced chip generations have smaller line widths (measured in nanometers), which translate into more transistors (i.e., logic gateways), more computing potential, and greater energy efficiency. In each quarter we identify the technology with the smallest line width then available and treat it as the *frontier* technology for its first year on the market, that is, for the quarter in which it arrives and the three that follow; thereafter it is no longer at the frontier. Because frontier status lapses after a year, there are quarters in which no technology is at the frontier: those in which the smallest line width available has held that position for more than a year and nothing more advanced has yet been introduced. A newly introduced “frontier” node is a poor substitute for the rest of the market, and the one-year window is meant to capture that novelty rather than a premium permanently attached to whichever node happens to be smallest. We denote by $\mathcal{F}_t$ the set of frontier technologies in quarter t , with $|\mathcal{F}_t| \leq 1$, and by $\mathcal{N}_t = K \setminus \mathcal{F}_t$ the set of non-frontier technologies, so that technologies are partitioned each quarter into two groups, $g \in \{F, N\}$.

At the lowest level of the nested CES structure, chips of a given technology k are aggregated across suppliers j :

$$Q_{kt} = \left[\sum_{j \in J} (\xi_{jkt} q_{jkt})^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

At the intermediate level, non-frontier technologies are aggregated into a single index:

$$Q_{Nt} = \left[\sum_{k \in \mathcal{N}_t} (\zeta_{kt} Q_{kt})^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}$$

while the frontier group contains a single technology, so that $Q_{Ft} \equiv Q_{kt}$ for $k \in \mathcal{F}_t$.

At the top level, the buyer substitutes between the two groups of frontier and non-frontier technologies:

$$Q_t = \left[\sum_g (\theta_{gt} Q_{gt})^{\frac{\iota-1}{\iota}} \right]^{\frac{\iota}{\iota-1}}$$

2. See Sweeting, Jia, Hui, and Yao (2022) and Deng, Jia, Leccese, and Sweeting (2024) for the case of strategic buyers.

where: q_{jkt} and Q_{kt} are the quantity purchased of product j and the quantity index for technology k respectively; ξ_{jkt} is product-level quality; ζ_{kt} is technology-node-level quality; θ_{gt} is group-level quality; σ is the within-technology elasticity of substitution; η is the across-technology elasticity of substitution; and ι is the across-group elasticity of substitution. We expect (but do not impose) that $\sigma > \eta > \iota > 1$, meaning that chips within a technology are more substitutable than chips across technologies, which are in turn more substitutable than frontier and non-frontier chips. Restricting frontier status to the first year of a node’s availability keeps ι interpretable as the elasticity of substitution between genuinely new chips and the rest of the market. This specification allows us to estimate a distinct price elasticity of demand for “frontier” products recognizing that demand for cutting-edge technologies may be considerably less price-sensitive because buyers place a high value on early access.

An outermost Cobb–Douglas layer over the semiconductor composite Q_t and an outside composite good closes the system by allocating a fixed share of aggregate expenditure to semiconductors.

We model product quality (ξ_{jkt}) as a function of observable product characteristics, and supplier fixed effects. Specifically, we model quality using the following functional form:

$$\xi_{jkt} = \bar{z}_{jkt}^\delta \times \nu_j^D \times \Delta \xi_{jkt}$$

where $\bar{z}_{jkt}$ denotes a “quality index” constructed on the basis of observable characteristics X_{jkt} , and whose construction is described in Appendix A.1. The characteristics included in X_{jkt} are the number of masks, metal layers, and polysilicon layers.³ The term ν_j^D denotes demand-side supplier fixed effects, and $\Delta \xi_{jkt}$ is the residual unobserved quality.

Utility maximization leads to a demand equation where demand for chips quantity q_{jkt} of supplier j ’s technology k in period t is a function of price p_{jkt} , the quality index $\bar{z}_{jkt}$, a set of supplier fixed effects, technology-time fixed effects interactions, and an idiosyncratic error term ε :

3. These are the key design and process choices that strongly influence both the quality and cost of a semiconductor device. A higher mask count generally reflects more complex circuitry and finer patterning, enabling greater transistor density and performance. Each additional mask, however, requires extra lithography, etching, and deposition steps, significantly raising manufacturing cost and cycle time. Similarly, increasing the number of metal layers expands the chip’s interconnect capacity, reducing signal congestion and enabling faster, more power-efficient designs. This adds to fabrication complexity, yield risk, and cost due to additional deposition and chemical-mechanical polishing stages. Polysilicon layers, typically used to form transistor gates, play a critical role in defining device performance and electrical characteristics; while the count is usually low (one or two), additional poly patterning can improve circuit functionality and adds modestly to cost.

$$\begin{aligned} \log(q_{jkt}) = & -\sigma \log(p_{jkt}) + \delta(\sigma - 1) \log(\bar{z}_{jkt}) + \underbrace{(\sigma - 1) \log(\nu_j^D)}_{\text{Supplier FEs}} \\ & + \underbrace{\log(R_{kt} P_{kt}^{\sigma-1})}_{\text{Technology/Time FEs}} + \underbrace{(\sigma - 1) \log(\Delta \xi_{jkt})}_{\text{Idiosyncratic shocks } (\varepsilon_{jkt}^l)} \end{aligned} \quad (1)$$

The term R_{kt} is total expenditure on technology k in period t while the term $P_{kt} = \left[\sum_j \xi_{jkt}^{\sigma-1} p_{jkt}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$ is the CES price index for technology k in period t .⁴

Similarly, we model technology quality (ζ_{kt}) as a function of technology fixed effects:

$$\zeta_{kt} = \phi_k^D \times \Delta \zeta_{kt}$$

where ϕ_k^D denotes demand-side technology fixed effects. The term $\Delta \zeta_{kt}$ is the residual unobserved quality. This specification allows for end-customer demand to vary across chip technologies.

The demand for technology $k \in \mathcal{N}_t$ is given by:

$$\log(Q_{kt}) = -\eta \log(P_{kt}) + \underbrace{\log(P_{Nt}^{\eta-1} R_{Nt})}_{\text{Time FEs}} + \underbrace{(\eta - 1) \log(\phi_k^D)}_{\text{Technology FEs}} + \underbrace{(\eta - 1) \log(\Delta \zeta_{kt})}_{\text{Shocks } (\varepsilon_{kt}^u)} \quad (2)$$

where $P_{Nt} = \left[\sum_{k \in \mathcal{N}_t} \zeta_{kt}^{\eta-1} P_{kt}^{1-\eta} \right]^{\frac{1}{1-\eta}}$ is the CES aggregate price index for non-frontier chips in quarter t and R_{Nt} is total expenditure on non-frontier chips.

Finally, we model group quality (θ_{gt}) as a function of time fixed effects:

$$\theta_{gt} = \rho_t^D \times \Delta \theta_{gt}$$

where ρ_t^D denotes demand-side time fixed effects and $\Delta \theta_{gt}$ is the residual unobserved quality. Allowing demand to vary across time for all suppliers, in all countries and technologies, accounts for aggregate demand shocks that impact the entire fabrication industry—for instance, as customers increase their demand for digital devices.

The demand for group g is given by:

$$\log(Q_{gt}) = -\iota \log(P_{gt}) + \underbrace{\log(P_t^{\iota-1} R_t)}_{\text{Time FEs}} + \underbrace{(\iota - 1) \log(\rho_t^D)}_{\text{Shocks } (\varepsilon_{gt}^u)} + (\iota - 1) \log(\Delta \theta_{gt}) \quad (3)$$

4. Though the technology-time fixed effects have a specific interpretation in the theoretical model, their interpretation in the empirical implementation of the demand model is broader as they also account for time-specific preference shifts towards a particular technology.

where $P_t = \left[\sum_g \theta_{gt}^{\epsilon-1} P_{gt}^{1-\epsilon} \right]^{\frac{1}{1-\epsilon}}$ is the CES price index in period t and R_t is total expenditure in period t . Note that, because the frontier group contains at most one technology, $Q_{Ft} \equiv Q_{kt}$ and $P_{Ft} \equiv P_{kt}$ for $k \in \mathcal{F}_t$.

3.3 Supply

As noted earlier, we assume that each country has only one supplier j (e.g., TSMC in the case of Taiwan, SMIC in the case of China). This assumption is driven by data constraints; in the data available to us, we can identify only the country in which a foundry is located, but not the foundry itself.

Cost Function

We model marginal costs as a function of the quality index $\bar{z}_{jkt}$, manufacturing wages in each supplier country, and a set of supplier, technology, and time fixed effects. The time fixed effects on the supply side capture among other things the prices of inputs that are internationally traded (e.g., silicon) and hence do not vary across countries. In addition, we follow Irwin and Klenow (1994) and assume period t marginal cost evolves with cumulative experience H_{jkt} :

$$c_{jkt} = \nu_j^S \times \phi_k^S \times \rho_t^S \times \bar{z}_{jkt}^\beta \times w_{jt}^\kappa \times H_{jkt}^\gamma \times e^{u_{jkt}} \quad (4)$$

$$H_{jkt} = 1 + Y_{jkt} + \underbrace{\alpha \times (Y_{jt} - Y_{jkt})}_{\text{Intra-Firm}} + \underbrace{\mu \times (Y_{Wkt} - Y_{jkt})}_{\text{Across Firms}} \quad (5)$$

where u_{jkt} is a mean-zero structural cost shock, w_{jt} denotes manufacturing wages in country j in period t , and ν_j^S , ϕ_k^S and ρ_t^S are supplier, technology, and time fixed effects, respectively. The set of fixed effects is similar to that on the demand side, but the rationale is different: Supplier fixed effects capture differences in marginal costs due to average (over time) subsidies; technology fixed effects account for the fact that different technologies may require different inputs or processes; and quarter fixed effects capture changes in costs across time as prices of internationally traded inputs change or input manufacturers change their prices either because they become more efficient (i.e., input prices fall) or because they gain pricing power.

Y_{jkt} is cumulative past output of technology k sold to all buyers ($Y_{jkt} = \sum_{\tau=0}^{t-1} q_{jk\tau}$) in previous periods. The terms Y_{jt} and Y_{Wkt} are the cumulative past output within the firm across the K technologies and the cumulative past output of technology k in the world, respectively. We refer to H_{jkt} as the “experience” of firm j producing technology k at period t and assume it is linear in past production. The parameter γ controls the importance of learning-by-doing as a function of a firm’s experience. When $\gamma = 0$ there exists no learning while for $\gamma < 0$, the firm’s cost decreases as it gains experience (i.e., as $H_{jkt} \uparrow$). When $\alpha > 0$, learning is

internalized by the firm across the K technologies so there exists economies of scope in the learning process. When $\mu > 0$, a firm's costs depend on experience accumulated by other countries producing the same technology. At this stage, we do not take a stand on the sources of this cross-border learning channel, but explore several mechanisms in the next section.

Firm Pricing

Each supplier maximizes expected discounted profits over its portfolio of technologies, denoted $K_j \subseteq K$, using quantity as the strategic variable. In Appendix A.4, we consider alternative assumptions regarding firm behavior (price competition; monopolistic competition; quantity competition with homogeneous products within technologies) and show that our results are robust to these alternative modeling assumptions.⁵

$$\max_{\{q_{jkt}\}} \left\{ E_0 \sum_{t=0}^{\infty} \left[\left(\frac{1}{1+d} \right)^t \sum_{k \in K_j} \left(p_{jkt} q_{jkt} - c_{jkt} q_{jkt} \right) \right] \right\}$$

The expectations operator E_0 denotes expectations as of the initial period; the supplier's period t quantity choice is conditioned on information available at t . d is a fixed exogenous discount rate, and c_{jkt} is the firm's marginal cost which varies across time because of learning-by-doing effects. The first-order condition shows that firms will set price as a markup over *dynamic* marginal cost:

$$p_{jk0} = \frac{\tilde{\varepsilon}_{jk0}}{\tilde{\varepsilon}_{jk0} - 1} \left(c_{jk0} + \overbrace{E_0 \left[\sum_{t=1}^{\infty} \sum_{r \in K_j} \left(\frac{1}{1+d} \right)^t \times \frac{\gamma \Delta_{r,k} c_{jrt} q_{jrt}}{H_{jrt}} \right]}^{\text{Dynamic Marginal Cost}} \right) \quad (6)$$

where $\Delta_{r,k} \equiv \begin{cases} 1 & \text{if } r = k \\ \alpha & \text{if } r \neq k \end{cases}$

Define Λ_{jkt} as the wedge between the static marginal cost equivalent implied by the contemporaneous product-market first-order condition and current marginal cost. Under quantity competition, this static marginal cost equivalent is marginal revenue, so $\Lambda_{jkt} \equiv \frac{\tilde{\varepsilon}_{jkt}-1}{\tilde{\varepsilon}_{jkt}} p_{jkt} - c_{jkt}$.

5. An alternative modeling strategy would be to assume that foundries and buyers negotiate over individual orders. It is not clear how restrictive our modeling approach is relative to this alternative. Modeling quantity choice or pricing as a Nash-in-Nash Bargaining problem would replace our assumption with one where we assume negotiations are done independently (Collard-Wexler, Gowrisankaran, and Lee, 2019) which is also strong. Furthermore, we would not be able to credibly identify the parameters of such a bargaining model given that we do not observe the identity of buyers.

Then (6) can be written in recursive form as:

$$\Lambda_{jkt} = \frac{1}{1+d} \times E_t \left[\Lambda_{jk,t+1} + \overbrace{\sum_{r \in K_j} \frac{\gamma \Delta_{r,k} c_{jr,t+1} q_{jr,t+1}}{H_{jr,t+1}}}^{\text{Dynamic Incentive}} \right] \quad (7)$$

where suppliers choose period t quantity conditional on information available at that time. We impose a zero continuation value beyond the end of the sample, so that the terminal-period first-order condition is static, i.e., the terminal condition is $p_{jkT} = \frac{\tilde{\varepsilon}_{jkT}}{\tilde{\varepsilon}_{jkT}-1} c_{jkT}$.

The variable $\tilde{\varepsilon}$ denotes the effective elasticity of demand facing the supplier, which will generally be a function of its market share within its technology, s_{jkt} , of the share of that technology within its group, s_{kt} , and—because foundries are multi-product—of the revenue shares of the supplier’s entire portfolio at each level of the nest. Note that this formulation is general and can accommodate several competition models popular in empirical work. In our model, which is based on the assumption of quantity competition, the supplier internalizes the effect of its quantity choices on every index it enters: its technology index Q_{kt} , its group index Q_{gt} , and the industry aggregate Q_t . Let $s_{gt} \equiv P_{gt} Q_{gt} / R_t$ denote group g ’s revenue share of total semiconductor expenditure (so that $s_{Ft} + s_{Nt} = 1$); $S_{jNt}^{-k} \equiv \frac{1}{R_{Nt}} \sum_{r \in K_j \cap \mathcal{N}_t, r \neq k} p_{jrt} q_{jrt}$ the revenue share of the supplier’s *other* non-frontier products in the non-frontier group; $S_{jNt} \equiv s_{jkt} s_{kt} + S_{jNt}^{-k}$ the supplier’s total revenue share of that group; and $S_{jt} \equiv \frac{1}{R_t} \sum_{r \in K_j} p_{jrt} q_{jrt}$ the supplier’s revenue share of total semiconductor expenditure. The perceived elasticity is then

$$\frac{1}{\tilde{\varepsilon}_{jkt}} = \begin{cases} \frac{1}{\sigma} + \left(\frac{1}{\eta} - \frac{1}{\sigma} \right) s_{jkt} + \left(\frac{1}{\iota} - \frac{1}{\eta} \right) S_{jNt} + \left(1 - \frac{1}{\iota} \right) S_{jt} & \text{if } k \in \mathcal{N}_t, \\ \frac{1}{\sigma} + \left(\frac{1}{\iota} - \frac{1}{\sigma} \right) s_{jkt} + \left(1 - \frac{1}{\iota} \right) S_{jt} & \text{if } k \in \mathcal{F}_t. \end{cases} \quad (8)$$

Since we expect $\iota < \eta$, this implies that, all else equal, markups on frontier chips will be higher than on non-frontier chips. We consider several alternative modes of competition in Appendix A.4, and show that our results regarding learning are robust to the mode of competition.⁶

From Equation 6, we observe that optimal price is a function of the standard markup and a dynamic component. Learning-by-doing makes the cost dynamic, so that the firm internalizes future cost reductions conditional on current period information. This is Boston Consulting’s famous “Experience Curve.” Importantly, the firm considers how technology k

6. The main difference across modes of competition is in the markups – as expected, quantity competition generates larger markups than price competition.

quantity choices impact the future marginal costs for all technologies (r) in its portfolio. We assume firms do not internalize the future competitive effects of their quantity decisions.⁷

If $\gamma=0$, there exists no scope for learning-by-doing and we get the standard markup over static marginal cost. As learning-by-doing effects grow ($\gamma \downarrow -\infty$), however, the supplier optimally chooses to decrease its price from the static optimum, leading to higher period t output and lower future costs. Similar intuition holds for $\alpha > 0$ as the supplier internalizes cross-technology experience leading it to decrease prices across all of its K_j technologies. When $\alpha = 0$, there is no cross-technology learning, so production of any other technology $r \neq k$ has no dynamic effect on technology k . The extent to which suppliers benefit from cross-country learning comes via μ . At this point, we remain agnostic about the sources of this learning channel and explore several potential mechanisms in the next section.

The strength of learning effects also depends upon the supplier's expectations about future demand and cost shocks via $E_t[c_{jr,t+1}q_{jr,t+1}]$. If the supplier thinks costs next period will be higher (i.e., high u), it chooses to decrease price today to increase production and reduce future costs by moving down the learning curve. On the other hand, if the supplier thinks there will be a positive demand shock next period, it reduces price to increase margins next period. These dynamic price incentives fall as the firm gains experience (i.e., $H \uparrow$), however.

3.4 Estimation and Identification

We recover estimates of the demand and supply parameters in two steps. First, we estimate the demand-side parameters $[\hat{\sigma}, \hat{\eta}, \hat{\iota}, \hat{\delta}, \hat{\nu}_j^D, \hat{\phi}_k^D, \hat{\rho}_t^D]$ using 2SLS (the instruments are described below). Given that buyers maximize static utility, the demand-side parameters are not affected by the learning parameters on the supply-side.

Next, we estimate the supply-side parameters $\hat{\theta} = [\hat{\gamma}, \hat{\alpha}, \hat{\mu}, \hat{\beta}, \hat{\kappa}, \hat{\nu}_j^S, \hat{\phi}_k^S, \hat{\rho}_t^S]$ using a generalized method of moments (GMM) estimator using the data of the GSA panel and the variables of period t , i.e.,

$$\hat{\theta} = \operatorname{argmin} \left\{ g(\theta)' Z^S W Z^{S'} g(\theta) \right\}, \quad (9)$$

where $g(\theta)$ is a stacked vector of the supply-side structural errors (u_{jkt}) and W is the weighting matrix. We use a one-step GMM estimator with a weighting matrix proportional to the inverse sample second moment of the instruments, $W = (Z^{S'} Z^S)^{-1}$. Inference is based on the heteroskedasticity-robust GMM covariance matrix. The GMM estimator exploits the fact that at the true value of the learning-by-doing parameters, $(\gamma^*, \alpha^*, \mu^*)$, the supply-side

7. This is done for tractability as estimating the strategic quantity or pricing decisions in dynamic games is complex due to multiple equilibria. See [Deng et al. \(2024\)](#) for a two-firm numerical example.

instruments Z^s are orthogonal to the structural errors because any ex-post errors are not correlated with ex-ante information available to the firms.

Recovering the underlying model parameters requires addressing price endogeneity in Equations (1)-(3) and unobserved expectation of future endogenous variables in Equation 6.

On the demand side, simultaneity bias arises from the potential correlation of prices with *unobserved* quality ($\Delta\xi_{jkt}$ in Equation 1, $\Delta\zeta_{kt}$ in Equation 2, and $\Delta\theta_{gt}$ in Equation 3). We remind the reader that we control for observable quality through product characteristics used to construct a “quality index” as well as a large set of fixed effects (supplier, technology, and time fixed effects). Our supply-side model suggests a natural instrument for addressing potential correlation between unobserved quality and price: a firm’s “experience” or stock of knowledge. The stock of knowledge affects price in our framework only through its effect on cost. Accordingly, we use a firm’s cumulative past output in technology k over the previous three years as well as its cumulative past output in all technologies over the previous three years, in the set of instruments for p_{jkt} . In addition, we include the real manufacturing wage in the supplier country in the demand-side instruments. In the estimation of the across-technology demand Equation 2, we construct similar instruments for the technology-level price index P_{kt} by averaging the aforementioned supplier-specific instruments across suppliers within a particular technology k . Finally, in the estimation of the across-group demand Equation 3, we construct instruments for the group-level price index P_{gt} by averaging the aforementioned instruments across the suppliers in each group.

On the supply-side, the instrument set includes the exogenous regressors and fixed effects that enter the cost equation, together with the following excluded instruments: a firm’s cumulative past output in technology k ($Y_{jkt} = \sum_{\tau=0}^{t-1} q_{jk\tau}$); a firm’s total cumulative past output in all technologies (Y_{jt}); the cumulative past world output in technology k (Y_{Wkt}), interacted with an indicator for line widths of 180nm and below; the one-period lag of a supplier’s log quantity sold and market share in technology k (capturing AR(1) demand forecasts); the inverse-distance⁸ of a particular chip technology k to all other chip technologies k' sold by the same supplier in the previous period, weighted by their market shares in the previous period; the number of quarters since the technology was introduced by any supplier times a dummy that takes the value one if the supplier is a follower and zero if the supplier was the first to introduce the technology; and the number of competitors in the previous period (i.e., number of rival products in the previous period in same technology), divided by the number of quarters the technology had been on the market at that time.⁹ These instruments

8. Distance is computed as the absolute value of the difference in line width of all pairwise chip technology combinations offered by the supplier, excluding the technology itself. Hence, the instrument takes the form: $\sum_{k' \in j, k' \neq k} \frac{s_{jk'}(t-1)}{1 + |Current_k - Previous_{k'}|}$.

9. In estimation, all experience-based instruments enter as $\log(1 + Y/10^3)$, where Y is measured in wafers (so that $(Y/10^3)$ is measured in thousands of wafers).

capture the fact that a firm’s stock of knowledge depends on the number of technologies it offers and the time each technology has been on the market as well as the order of entry; late entrants benefit more from a larger stock of accumulated knowledge. Note that we do not model entry in this paper; however, we use entry – the order of entry in particular – to identify the learning parameters. Finally, we set the quarterly discount rate to $d = 4\%$, corresponding to an annualized rate of roughly 17%, in line with the cost of capital faced by firms in this capital-intensive and cyclical industry.

The estimation algorithm is described in detail in Appendix A.2.

3.5 Data

The data used to estimate the model are from a proprietary database collected by the Global Semiconductor Alliance (GSA). The GSA is a nonprofit semiconductor industry association spanning the semiconductor ecosystem. Its wafer-pricing survey collects confidential purchasing data from fabless firms and IDMs that source wafers from foundries. Each quarter the organization collects information on wafer fabrication prices, quantities, and characteristics of their orders from both domestic and foreign foundries. Responses are anonymous and firms which participate are granted access to the results. The dataset consists of 17,089 individual quarterly responses to the “Wafer Fabrication & Back-End Pricing Survey” covering years 2004-2019.¹⁰ According to GSA the sample is representative of the industry and accounts for roughly one-fifth of all fabless semiconductor wafers produced worldwide. Our analysis focuses on microprocessors, “system-on-chip,” and commodity ICs. We exclude memory chips because these are manufactured by large technological conglomerates such as IBM and Samsung, which also supply a large set of products outside the industry that are subject to different economic forces.

The data include nominal price paid, the number of wafers purchased, and the foundry’s country location. We also observe wafer characteristics, including the line width, wafer size, and number of layers. We can therefore examine how foundry wafer prices vary by foundry location, after controlling for physical characteristics. Recall, we cannot identify the specific foundry which fulfilled each order, though the dominance of Taiwan Semiconductor Manufacturing Corporation (TSMC) in the Taiwanese market, Semiconductor Manufacturing International Corporation (SMIC) in Chinese market, and Chartered Semiconductor in Singapore market suggests that transactions which involve wafers fabricated in the Taiwan, Chinese, and Singapore markets were fulfilled by these firms, respectively. U.S. foundries produced on average 7% of the wafers during the sample. In comparison, fabrication plants in Taiwan, Singapore, and China accounted for 56%, 8%, and 10% of the market, respectively. Thus, while there are a variety of foundries producing semiconductor wafers for fabless

10. Although data up to 2021 exist, we exclude observations from the COVID era.

semiconductor firms, the bulk of the wafers are produced overseas in Taiwan, presumably by TSMC.

3.5.1 Descriptive Analysis: Pricing Trends

Figure 3 plots wafer prices for two different technologies, 90 nm and “mainstream” (50-89 nm),¹¹ where the number refers to line width. Smaller line widths correspond to more advanced technologies. The plots suggest that prices start high and fall over time even when there is little competition. This downward trend is suggestive of learning-by-doing specific to each technology.

An alternative explanation for the decline in prices is increased competition rather than falling costs. As more suppliers enter production, markups, and therefore prices, may fall. Our empirical model allows for this possibility. On the supply side, both the quantity- and price-competition specifications imply that markups decline as additional suppliers enter the market. The learning parameters we estimate are therefore net of this competitive effect.

Another interpretation, based on the demand side, is that in the early years after a new technology is introduced, demand may be especially strong because close substitutes are not yet available, which can lead to high prices. We capture this mechanism through a distinct price elasticity of demand for “frontier” technologies as well as through time–technology interactions in the demand specification.¹² The learning parameters we estimate are again net of this demand-side effect.

The figure also helps clarify the various dimensions of learning and our identification strategy.

Consider the price trajectories in Panel (a) for the 90 nm technology. For clarity, we focus on three suppliers: two early innovators (Taiwan and Japan) and a follower (China). A first observation is that China enters the market at a significantly lower price than the two leaders. A second observation is that prices for Taiwan and Japan fall rapidly in the early years after entry. Prices for China also decline over time, although the slope is much flatter (the scale of the figure makes this less visible).

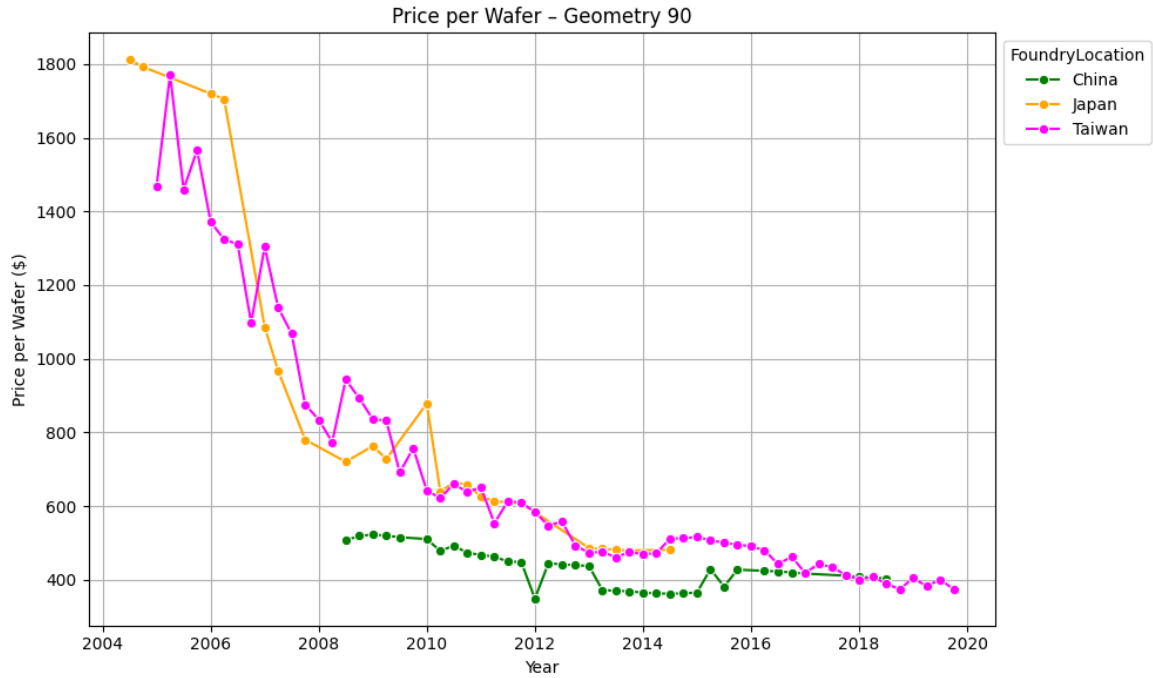
If we were to estimate the learning rate using the price paths of these three suppliers without differentiating between within-supplier versus across-supplier spillovers that followers enjoy, we would likely obtain a small learning effect. In contrast, if we allow late entrants (such as China) to benefit from learning generated by earlier entrants, we would attribute (at least part of) the lower entry price and the slower post-entry price decline to cross-supplier learning. We use these price differences at the time of entry to identify cross-border learning.

11. The term “mainstream” corresponds to industry jargon.

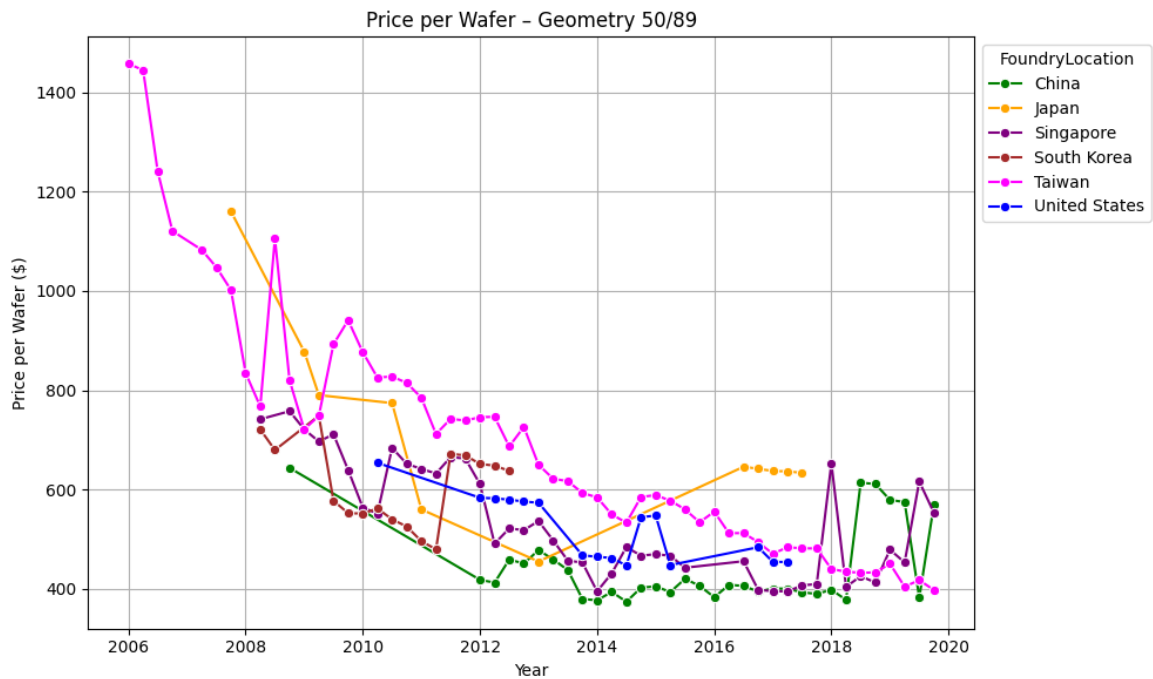
12. In addition, the effect of lack of (or little) competition is captured through the supply-side competition model that allows for markups to evolve with the number and market shares of competitors.

Figure 3: Pricing Trends

(a) Technology: 90nm



(b) Technology: “Mainstream” (50–89 nm)



The lower prices offered by China could also reflect lower quality or a cost advantage. Given that our demand and cost specifications control for observable quality characteristics

and time-invariant cross-country differences, while the IV strategy addresses remaining unobserved quality correlated with price, the cost-advantage explanation appears more plausible. Such an advantage could arise from lower wages, more efficient production practices, or government subsidies. The descriptive analysis alone cannot distinguish among these possibilities. In our estimation, however, we control for cross-country wage differences, while time-invariant subsidies are absorbed by country fixed effects in the cost function.

A remaining possibility is that China subsidizes the early stage of production until learning-by-doing takes hold, allowing firms to become cost-competitive over time. While we cannot rule this out, the pattern we observe is not unique to China or to the 90 nm technology. Panel (b) shows similar price paths for the “mainstream” technology across several other supplier countries. In that panel as well, all follower countries enter at lower prices than the two leaders (Taiwan and Japan). This pattern forms the basis of our identification of cross-border learning.

3.6 Empirical Results

3.6.1 Results: Key model estimates

This subsection discusses some key estimates and their implications for policy. The full set of estimation results are provided in Appendix A.3.

Demand-side Estimates The demand-side estimates are displayed in Table A.1 in Appendix A.3. The three elasticities of substitution are large and precisely estimated, with magnitudes that conform to our expectations: The lower-nest elasticity of substitution $\sigma = 16.36$ (*s.e.* = 1.58) is larger in absolute value than the middle-nest elasticity of substitution $\eta = 10.01$ (*s.e.* = 1.90), which is in turn larger in absolute value than the upper-nest elasticity of substitution $\iota = 4.68$ (*s.e.* = 0.29). The parameter estimate for the “quality index” is positive and precisely estimated: 4.32 (*s.e.* = 0.97). As noted earlier, these estimates are not affected by the supply-side specification given that buyers are assumed to be static utility-maximizers.

Supply-side Estimates The supply-side estimates are reported in Table A.2 in Appendix A.3. We estimate three specifications. In the first, learning-by-doing operates only at the supplier–technology level (1). In the second, we additionally allow for learning across technologies within a supplier (2). In the third, we also allow for cross-border learning across suppliers (3). Specification (3) is our baseline, and we base the discussion on this specification. We nonetheless report the first two specifications, as they help clarify the nature of learning and the sources of identification in our setting.

All three specifications control for cost differences across technologies as well as sources of cost differences across countries. Labor cost is a significant cost component in semi-

conductor manufacturing, and our specification controls for wages. The second significant component of variable costs is the cost of inputs; these are internationally traded and absorbed by the time fixed effects. The empirical specifications also control for quality differences across suppliers that are associated with higher costs.

The quality index enters with a positive and statistically significant coefficient in all specifications, while the wage coefficient is small and statistically insignificant. The production-side fixed effects are not reported for ease of presentation.

The estimated Spence learning rates under different assumptions about learning spillovers are reported in the last row of Table A.2 in Appendix A.3. When we allow learning only at the supplier–technology level, and rule out both within-foundry learning across technologies and cross-border learning, the estimated learning rate is small, at 3.7%, with a 95% confidence interval of [2.9%–4.4%]. This estimate is well below industry lore and prior evidence, such as Irwin and Klenow (1994), who report learning rates in the range of 20–30%.¹³

The pricing patterns in Figure 3 help explain why this estimate is so low. As discussed in the descriptive analysis, prices for Taiwan and Japan—the two technology leaders—fall rapidly in the early years. In contrast, prices for the follower country, China, decline much more slowly over time. Because we identify learning-by-doing from price changes over time, the relatively flat price path for China in Figure 3 leads to a low estimated learning rate when cross-border effects are not allowed.

Once we allow for learning across technologies within a foundry, the estimated learning rate rises to 11.8%, with a 95% confidence interval of [6.2%–17.0%]. This estimate is consistent with anecdotal evidence pointing to economies of scope in learning. For example, semiconductor firms often foster a culture of experimentation and innovation (see Section 2), which is likely to benefit multiple product lines and facilitate the introduction of newer, more advanced technologies. In specification (2), learning across technologies is identified from variation in a supplier’s cumulative experience across all technologies, using lagged firm-wide experience, together with lagged quantity and same-technology experience, as instruments. The number and timing of technologies produced by a supplier generate variation in this firm-wide experience stock.

When we additionally allow for cross-border learning, the estimated learning rate increases further to 17.3%, with a 95% confidence interval of [9.5%–24.4%]. The intuition for this result is again illustrated in Figure 3. In the 90 nm technology, China enters at a lower price than the incumbents, Taiwan and Japan. In our framework, this lower entry price partly reflects learning from the earlier experience of other countries. As discussed above, this pattern is not specific to the 90 nm technology but also appears in other technologies and for all follower countries in our sample.

13. Irwin and Klenow (1994) study DRAM memory chips, which are much more homogeneous than the integrated circuits considered here.

Focusing on the baseline specification in Column 3, which allows for all three types of learning, the estimates indicate that learning at the supplier–technology level is the strongest channel. The estimated values of γ , α , and μ are -0.27, 0.14, and 0.12, respectively. These estimates imply that a foundry learns about eight times more from an additional unit of its own cumulative output in a given technology than from an additional unit of cumulative output in that technology produced by foundries in other countries.

At the same time, the cumulative past output of other foundries worldwide is large — often more than eight times the cumulative output of a single foundry in that technology. As a result, cross-border learning is substantial and, in aggregate, it likely outweighs the contribution of a foundry’s own cumulative output. This type of learning is particularly important for follower countries, which – by definition – enter production with zero own cumulative output but can benefit from the accumulated experience of earlier entrants. In contrast, innovators (that is, countries that enter first, such as Taiwan) cannot draw on knowledge from earlier entrants in that technology, although they can still benefit from learning accumulated through the production of other technologies via economies of scope.

In sum, our results point to the significance of learning-from-experience, with a sizable cross-border component. The sources of the large cross-border learning estimated in our framework are the focus of section 4. One possible explanation is learning along the global supply chain. In particular, buyers – the fabless firms in our data – as well as input suppliers may play a key role in spreading knowledge across borders. To leverage economies of scale in chip manufacturing, chip designers must coordinate closely with foundries and input suppliers (such as ASML) on the exact specifications of the chips and the manufacturing process. Once these specifications are established, other chip designers can adopt them to benefit from manufacturing economies of scale. This leads to the standardization of designs and processes and allows foundries to pool orders from multiple buyers. Once designs and processes are standardized, buyers and input suppliers can approach foundries in other countries with potential orders, sharing information gained from earlier interactions with different foundries. This process can create significant spillovers not only across buyers but also across foundries. For example, if TSMC works with fabless firms and ASML to improve a feature of the production process, that knowledge may diffuse through ASML’s interactions with other foundries, generating spillovers throughout the industry.

Other mechanisms that could generate cross-border learning are foreign direct investment (FDI), R&D collaborations, technology licensing and/or the recruitment of engineers or other experts from foundries in other countries. We explore some of these channels in the next section.

3.6.2 Implications of Cross-border Learning for Policy

We interpret the magnitude of the estimated cross-border learning effects and their policy implications through two counterfactual exercises. In the first, we examine the consequences of restricting a particular country’s access to cross-border learning related to advanced technologies. In the second, we study the effects of policies that support domestic semiconductor manufacturing in a particular country. In both counterfactuals we hold fixed product availability, demand shifters, and aggregate semiconductor expenditure at their benchmark paths, while allowing prices, quantities, experience stocks, marginal costs, and markups to adjust endogenously.

We do not view these counterfactuals as literal representations of real-world policy scenarios. Actual policies are considerably more complex, and our model is too stylized to capture all of their relevant dimensions. Nevertheless, as we discuss below, several recently implemented policies contain elements that closely resemble the interventions considered in our counterfactuals. We therefore use these exercises primarily to illustrate the economic mechanisms that arise in the presence of cross-border learning and to show how such learning can alter the effects of policy interventions.

Restricting Cross-Border Learning In our first counterfactual exercise, we restrict China’s cross-border learning in advanced semiconductor technologies. In our data, we classify technologies with line widths between 11 and 49 nm as advanced. We consider two cases. In the first, more extreme case, we set $\mu = 0$ for China and for these technologies only, thus shutting down cross-border learning for Chinese production in the 11–49 nm range while leaving all other learning channels unchanged. Although a complete elimination of cross-border learning is unlikely to be feasible in practice, this case provides a useful benchmark for assessing the quantitative importance of the mechanism. In the second case, we reduce μ by half, again only for China and for technologies in the 11–49 nm range. This counterfactual captures a substantial, but incomplete, restriction on China’s ability to benefit from learning from other countries.

The second case is closer in spirit to the targeted technology restrictions adopted by the United States beginning in 2022, which sought to limit China’s access to advanced semiconductors, semiconductor-manufacturing equipment, software, and technical expertise. These measures exemplify what was described as a “small yard, high fence” approach: imposing stringent restrictions on a relatively narrow set of technologies considered critical to national security rather than pursuing broad technological decoupling (Sullivan, 2023; U.S. Department of Commerce, Bureau of Industry and Security, 2022). More broadly, these policies illustrate the potential for “weaponized interdependence,” whereby countries exploit asymmetric positions in global production and technology networks to restrict other countries’ access to critical inputs and knowledge (Farrell and Newman, 2019b; Beaumier and Cartwright,

2024). In semiconductors, this leverage is particularly salient because key segments of the supply chain are highly concentrated geographically and technologically (see our related discussion in Section 2). U.S. export controls, together with complementary restrictions adopted by the Netherlands, have limited China’s access to advanced semiconductor manufacturing equipment, including advanced lithography systems produced by ASML. Restrictions on U.S. persons’ support for certain advanced semiconductor facilities in China and, more recently, restrictions on U.S. outbound investment in Chinese semiconductor and microelectronics activities provide additional channels through which access to foreign expertise, capital, and knowledge may be curtailed (U.S. Department of Commerce, Bureau of Industry and Security, 2022; Government of the Netherlands, 2023; U.S. Department of the Treasury, 2024).

The main results are displayed in Table 1, while detailed results by country are shown in Appendix A.5, Tables A.4 and A.5. They yield several insights.

First, cross-border learning appears quantitatively crucial for China’s ability to compete in the 11-49 range. In the model, marginal cost is decreasing in experience because $\gamma < 0$. Cutting China’s μ therefore removes part of the cost advantage China obtains from accumulated production experience elsewhere in the world. The resulting cost increase translates into higher equilibrium prices and, because buyers can substitute toward competing producers, a much larger decline in Chinese quantities.

Second, the effects are highly nonlinear. Going from half- μ to zero does considerably more damage than a simple proportional extrapolation would suggest: the price effect rises from 5.7% to 13.3%, while the quantity contraction rises from 58.8% to 86.7%. This reflects the interaction between lost learning, declining production, weaker subsequent learning from experience, and equilibrium substitution toward competitors.

Third, the effect on the rest of the world (ROW) is very small in aggregate, even though individual producers – notably Japan and the United States – gain significantly in 11–49 quantities. One reason is that China’s baseline 11–49 production is quite small relative to ROW production. Hence even an enormous percentage contraction in Chinese production need not produce a large percentage change in aggregate world production outside China. There is also substantial redistribution within the ROW rather than a uniform expansion of all foreign producers.

Finally, the 50–89 results suggest cross-technology substitution rather than a major propagation of the shock. China’s 50–89 output rises modestly when its 11–49 learning is restricted, while prices barely move. This is consistent with demand and production shifting toward the older technology as Chinese 11–49 becomes less competitive. The corresponding effects outside China are extremely small.

For comparison, we repeat the exercise above for Taiwan, i.e., setting the μ parameter for Taiwan and for the 11–49 nm technology only equal to zero or to half its estimated value.

Table 1: Restricting China's Cross-Border Learning

	Mean ΔP (%)	Median ΔP (%)	Aggregate ΔQ (%)
Panel A: $\mu_{\text{China}, 11/49} = 0$			
<i>11–49 nm</i>			
China	13.32	13.37	-86.68
Rest of world	0.04	0.00	-0.12
<i>50–89 nm</i>			
China	-0.02	0.01	0.63
Rest of world	-0.00	-0.00	0.03
Panel B: $\mu_{\text{China}, 11/49} = \hat{\mu}/2$			
<i>11–49 nm</i>			
China	5.70	5.73	-58.80
Rest of world	0.03	0.01	-0.10
<i>50–89 nm</i>			
China	-0.01	0.01	0.45
Rest of world	-0.00	-0.00	0.05

Notes: The table reports equilibrium changes relative to the benchmark. In Panel A, China's cross-border learning parameter μ is set to zero for 11–49 nm technologies only. In Panel B, it is set to one-half of its estimated value, $\hat{\mu}/2$, again for China and 11–49 nm technologies only. All other parameters are held at their benchmark values. “Rest of world” includes all countries in our sample other than China. Mean and median price changes are calculated across product-quarter observations. Aggregate quantity changes are computed from total quantities within each region and technology.

Although this scenario does not correspond to any policy contemplated in practice, it provides a useful illustration of the underlying mechanism.

Table 2: Restricting Taiwan's Access to Cross-Border Learning

	Mean ΔP (%)	Median ΔP (%)	Aggregate ΔQ (%)
Panel A: $\mu_{\text{Taiwan}, 11/49} = 0$			
<i>11–49 nm</i>			
Taiwan	0.3839	0.4536	-2.7430
Rest of world	0.1996	0.2069	-1.3282
<i>50–89 nm</i>			
Taiwan	-0.0828	-0.0827	0.7088
Rest of world	-0.0602	-0.0601	0.4734
Panel B: $\mu_{\text{Taiwan}, 11/49} = \hat{\mu}/2$			
<i>11–49 nm</i>			
Taiwan	0.1941	0.2299	-1.3972
Rest of world	0.1033	0.1049	-0.7107
<i>50–89 nm</i>			
Taiwan	-0.0427	-0.0421	0.3658
Rest of world	-0.0319	-0.0319	0.2575

Notes: The table reports equilibrium percentage changes relative to the benchmark. In Panel A, Taiwan's cross-border learning parameter μ is set to zero for the 11–49 nm technology only. In Panel B, it is set to one-half of its estimated value, $\hat{\mu}/2$, again only for Taiwan's 11–49 nm technology. All other country-technology learning parameters remain at their benchmark values. "Rest of world" includes all countries other than Taiwan. The effects on 50–89 nm are indirect equilibrium reallocation effects.

The comparison with the China experiment reveals that Taiwan itself is far less sensitive to the loss of cross-border learning. Eliminating μ for Taiwan 11–49 raises its prices by only about 0.38% and reduces quantity by 2.74%, compared with the very large effects we found for China. This is consistent with Taiwan's much deeper accumulated own production experience making foreign experience a smaller marginal component of its effective knowledge stock. At the same time, the general-equilibrium effects are much broader because Taiwan is an enormous 11–49 producer relative to the rest of the world. Consequently, most other 11–49 producers also experience higher prices and lower quantities.

This comparison is informative in that it demonstrates that access to foreign knowledge is substantially more important to follower countries, such as China, than to technology leaders, such as Taiwan.

Supporting Domestic Manufacturing A second counterfactual exercise examines the effects of supporting U.S. semiconductor manufacturing through policies that lower domestic production costs. Specifically, starting in 2015Q1,¹⁴ we consider a permanent 10 percent reduction in the non-learning cost shifter of U.S. foundries, which can be interpreted as arising from subsidies or other forms of government support. As in the previous exercise, we do not view this counterfactual as a literal representation of any particular policy. It nevertheless captures an important element of the CHIPS and Science Act, which provided substantial incentives for expanding semiconductor manufacturing capacity in the United States. Our counterfactual abstracts from the entry of new facilities and instead isolates the effect of lowering production costs for domestic producers.

Our main interest is whether such support can be effective when a country has fallen substantially behind the technological frontier in manufacturing. Although the United States remains a global leader in semiconductor innovation, design, and other high-value segments of the industry, its share of global semiconductor manufacturing capacity has declined sharply over the past several decades. The previous counterfactual suggests that cross-border learning is particularly important for follower countries, raising the possibility that the effectiveness of domestic production subsidies depends critically on continued access to foreign knowledge. We therefore hypothesize that domestic support can generate substantial gains when U.S. producers remain connected to international sources of knowledge, but will be considerably less effective when those learning channels are disrupted.

To investigate this interaction, we conduct the counterfactual under two alternative assumptions. In the first, U.S. foundries retain access to cross-border learning. In the second, we shut down the cross-border learning channel for U.S. producers. Comparing the two cases allows us to assess whether domestic production support can substitute for access to foreign knowledge, or whether the two are instead complements.

The results for the U.S. are displayed in Table 3. Additional results for the rest of the world are shown in Appendix A.5, Table A.6.

The main insight conveyed by Table 3 is that domestic industry support and cross-border learning are complements. With cross-border learning intact, the subsidy lowers U.S. prices in every technology observed after 2015 and raises U.S. output in every technology. Once cross-border learning is removed, that pattern disappears. Prices rise substantially in the 130/140

14. We chose this starting date because by then, most technologies in our sample had entered the market, allowing us to capture rich cross-technology substitution while retaining four full years to trace the cumulative effects of the policy changes.

Table 3: US Effects of a 10% Production Subsidy With and Without Cross-Border Learning

Technology	Mean ΔP (%)		Aggregate ΔQ (%)	
	Subsidy only	Subsidy + $\mu_{US} = 0$	Subsidy only	Subsidy + $\mu_{US} = 0$
11/49	-5.14	2.67	14.89	8.74
50/89	-4.93	1.12	24.87	32.20
90	-5.55	-2.85	26.21	94.37
130/140	-5.75	16.60	78.09	-86.75
150/180	-7.69	13.47	74.94	-67.15
250/350	-4.32	1.02	63.09	-19.86
400/600	-4.59	-5.69	8.99	209.94

Notes: The table reports equilibrium percentage changes relative to the benchmark for US semiconductor production from 2015Q1 onward. “Subsidy only” denotes a permanent 10% reduction in the US cost shifter beginning in 2015Q1, with the cross-border learning parameter μ_{US} remaining at its benchmark value. “Subsidy + $\mu_{US} = 0$ ” denotes the same subsidy with US cross-border learning set to zero for all technologies beginning in 2015Q1. Mean price changes are unweighted across product-quarter observations. Aggregate quantity changes are computed using summed quantities within each technology.

and 150/180 technologies and output collapses. The same fiscal support therefore has very different effects depending on whether U.S. producers can continue to draw on knowledge accumulated abroad. A subsidy can expand domestic production when producers continue to learn from the world, but the same subsidy may be insufficient to offset the cost increase caused by technological isolation. That said, the importance of cross-border learning is highly technology-specific. The response is not uniform across technologies because the composition of knowledge differs across them. Technologies in which foreign accumulated experience is a large component of the U.S. knowledge stock are hit hard when cross-border learning is shut down. Technologies that rely relatively more on own-technology or within-foundry experience suffer much less. This heterogeneity helps explain why 130/140 and 150/180 contract sharply while 400/600 does not. The exceptionally large expansion of 400/600 is further amplified by equilibrium reallocation: contractions in other U.S. technologies reduce the U.S. overall multiproduct market power and hence its markup in 400/600, while demand shifts toward that technology.

Foreign producers are affected through equilibrium forces, but the largest effects remain domestic. Under the subsidy-only counterfactual, ROW prices generally fall because stronger U.S. competition puts downward pressure on equilibrium prices (see Table A.6). When U.S. cross-border learning is eliminated, the sign reverses: ROW prices rise modestly in most technologies as U.S. competitive pressure weakens. The foreign quantity changes are

generally much smaller than the large reallocations across U.S. technologies, highlighting that the primary adjustment margin in this experiment is the composition of U.S. production.

As emphasized above, we do not interpret these counterfactuals as literal representations of real-world policy scenarios. They nevertheless provide useful insights for policy. A common criticism of the CHIPS Act was that the U.S. had little prospect of catching up with producers in countries such as Taiwan, which had accumulated far more manufacturing experience. Our final counterfactual highlights an important limitation of this argument. It would be compelling if U.S. producers had to rely solely on their own accumulated experience. With access to foreign production experience, however, the prospects for successful catch-up are substantially greater. Such access can take many forms, including recruiting experienced engineers from TSMC and other foreign producers, collaborating with TSMC and other frontier manufacturers, and drawing on the expertise of equipment suppliers such as ASML and other firms throughout the semiconductor supply chain.

4 Mechanisms

The cross-border learning term estimated in the previous section captures several distinct mechanisms through which knowledge generated in one country becomes available to producers in another. In this section, we examine three specific channels that, based on the industry evidence discussed in Section 2, appear particularly important for cross-border knowledge transmission: research collaborations along the supply chain, labor flows, and voluntary knowledge sharing by firms. Because some of these channels involve tacit or proprietary knowledge, they leave limited direct evidence, and therefore, the methods we employ are eclectic - a combination of purely descriptive and historical evidence with reduced form analysis based on different data sources. Nevertheless, we believe that, taken together, the evidence supports the main thesis of our paper: the cross-border learning effect we estimate is consistent with an important role for market-mediated relationships rather than exclusively ambient technological spillovers. This suggests that at least part of the cross-border learning channel is policy-manipulable: firms can internalize it, and governments can restrict access to it.

4.1 R&D collaboration along the supply chain

Section 2 argued that collaboration in R&D along the supply chain is necessary given the technological interdependencies inherent to advancing technology in the industry and provided illustrative examples. We now turn to examining collaboration patterns systematically. We are interested in two related questions. First, is R&D collaboration as common as the narrative

evidence suggests? Second, if so, what structure do R&D collaborations take? Specifically, does the network of observed collaborations imply that knowledge can be transmitted across foundries, including across countries, through market-mediated transactions with suppliers and buyers?

We examine these questions using three complementary approaches. First, we examine R&D collaboration as proxied by co-authoring among engineers employed at different semiconductor firms building on Kapoor and McGrath (2014). This measure is the most granular of the three, as it captures collaboration at the level of individual engineers located at a particular establishment (of a potentially multi-establishment, multinational firm). The main limitation of this measure is that it does not reveal the technological or economic value of the collaborations we observe to the firms involved. Thus, we add a second measure by extracting mentions of R&D collaboration from publicly listed firms' earnings calls. The idea behind this second measure is that senior management arguably use scarce time during a quarterly call to report on technological bottlenecks that materially impact the firm's profitability. Third, to inform the question of whether this knowledge can be transmitted to other customers of the firm by one of the collaborating parties, or whether the knowledge is jointly protected by intellectual property rights (IPR), we examine co-patenting patterns in the industry. The data sources and measurement procedures for each of these measures are described in detail in Appendix B.1 and in the notes to Table 4. We emphasize here, however, that our selection criteria are designed to capture collaborations involving practical, implementation-related knowledge rather than basic research. For example, the technology areas used to identify relevant co-authorship, listed in the notes to Table 4, are chosen with this distinction in mind.

R&D collaboration in the engineering literature

We begin with the first question: how common is cross-firm R&D collaboration? Table 4 reports the results. Because the four venues publish academic and industrial research side by side, most of the 7,794 papers in the corpus have no company-affiliated author (column 1). Our measures are therefore defined on the 2,287 papers with at least one company-affiliated author (column 2) as this is the population within which collaboration between firms can be observed.

Our central result is that the scale of cross-firm co-authorship is striking: between one in five and one in three of the company-authored papers have authors at two or more distinct firms (column 3). It also frequently crosses borders: 10 to 14 percent of company-authored papers involve two or more firms that could not have collaborated within a single country (column 4) — roughly 40-50 percent of all cross-firm papers. A smaller share, 3 to 5 percent, are papers from a single company whose authors are nonetheless located in two or more countries (column 5), a first indication of the intra-firm multinational channel we return to

Table 4: R&D Collaboration in the Frontier Semiconductor Engineering Literature

Corpus	All papers (1)	Company-authored (2)	Share of company-authored papers (%)		
			Cross-firm (3)	Cross-firm, cross-border (4)	Sole company, cross-border (5)
Baseline	7,794	2,287	22.3	10.4	4.3
Excl. design enab.	4,209	1,367	26.8	13.7	4.5
Excl. design enab. and EUV	1,393	262	29.8	11.5	3.4

Notes: The table reports co-authorship patterns in the core technical journals of semiconductor process engineering. The corpus consists of all papers published 2010–2025 in four venues: the SPIE conference proceedings, the *IEEE Transactions on Semiconductor Manufacturing*, the *Journal of Micro/Nanolithography*, and the *IEEE Transactions on Computer-Aided Design*, matching one of twelve frontier technology areas fixed prior to the search: EUV lithography, EUV sources, EUV resists, pellicles, mask 3D effects, stochastic defects, high-NA lithography, design–technology co-optimization, system–technology co-optimization, process design kits, gate-all-around, and design enablement. Papers and author affiliations are from OpenAlex (retrieved July 2026). The second row excludes the design-enablement area; the third additionally excludes the seven EUV-related areas, leaving the four design- and device-side areas. Column (1) counts all papers, including those with academic and research-institute authors only. Column (2) counts papers with at least one author affiliated with a company. Firm names are normalized so that a company’s establishments in different countries count as a single firm. Column (3) is the share of company-authored papers whose company affiliations include two or more distinct firms, our benchmark measure of cross-firm collaboration. Column (4) is the share of company-authored papers with two or more firms that are cross-border under the following rule: no single country hosts establishments of all firms on the paper, so the collaboration could not have taken place within one country. Column (5) is the share of company-authored papers with a single firm whose establishments on the paper span two or more countries. Throughout, an affiliation’s country is the location of the establishment, not of the firm’s headquarters.

below. The pattern is not an artifact of the largest or the most familiar technology areas: excluding design enablement, and then the seven EUV-related areas, raises the cross-firm share, and the cross-border share remains stable.

Taken together, these results show that firms in this industry work together on R&D far more than an arm's-length relationship would imply, and that this collaboration routinely spans countries. The narrative evidence in Section 2 establishes that co-authorship of this kind reflects joint development work rather than courtesy attribution. Next, we turn to characterizing the structure of collaboration.

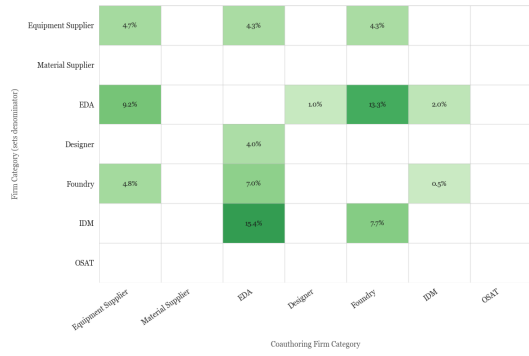
The network of R&D collaboration

While R&D collaboration among firms may be quite common in the industry, to understand whether knowledge jointly created by firms can be voluntarily shared across foundries by firms' buyers and suppliers, we also need to understand what form this collaboration takes. To this end, we study the network of collaboration between leading firms at different parts of the value chain. Specifically, for each step of the value chain in Figure 4, we select three leading firms and study R&D collaboration using all three measures.¹⁵ These firms are located in different countries and, as emphasized in Section 2, different stages of the semiconductor supply chain are concentrated in different countries. Collaboration along the supply chain therefore inherently entails cross-border collaboration.

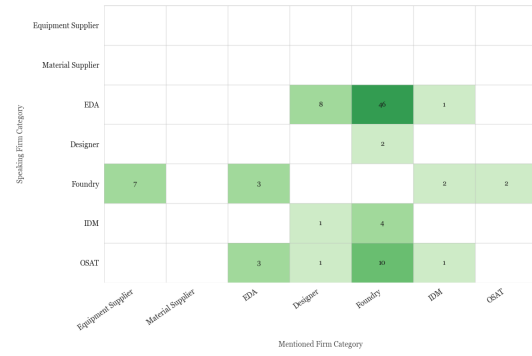
Figure 4 displays cross-firm collaboration by position in the value chain: panel A the co-authorship shares from the engineering literature, panel B the count of R&D collaborations disclosed in earnings calls. Both sources display the same central finding: R&D collaboration in the industry occurs vertically, between buyers and suppliers. Across both sources, EDA firms in particular are central in the network of collaboration working with foundries, designers, equipment suppliers and OSAT firms. That the design automation tier should be central is what one would expect based on our discussion in Section 2. The interface between design and manufacture is where technological interdependence is most acute, and where it is most readily resolved by working together: design tools must encode the manufacturing process they target.

In contrast, the diagonals of the matrix, where direct competitors sit side-by-side, are rarely populated. Collaboration is organized around complementarity rather than substitution, and the empty cells are correspondingly those linking firms that sell the same thing. No foundries are working together, nor are any EDA firms. Equipment suppliers (and material suppliers) are the exception, and for good reason as these firms occupy different steps of the

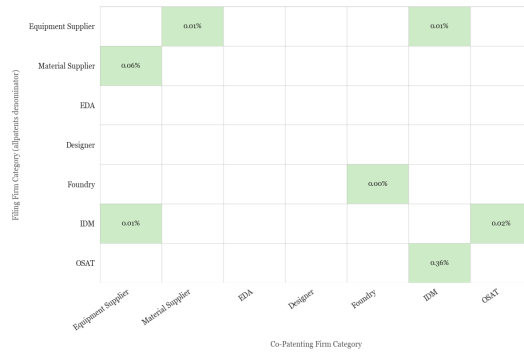
15. The leading firms are: ASML, Applied Materials, Tokyo Electron Ltd. (equipment suppliers), Dow Chemical, Du Pont, Showa Denko (materials suppliers), Cadence, Mentor Graphics, Synopsys (EDA firms), Broadcom, Nvidia, Qualcomm (designers), GlobalFoundries, TSMC, UMC (foundries), SK Hynix, NXP, Texas Instruments (IDMs), ASE, Amkor, UTAC (OSAT).



(a) Share of papers with cross-firm co-authors



(b) Count of R&D collab. in earnings calls



(c) Share of co-patenting

Figure 4: Matrix of R&D collaborations

Notes: Each panel reports R&D collaboration among the 21 leading firms listed in footnote 15, three per value-chain segment, over 2010–2025. Diagonal cells pair distinct firms within the same segment. Shading is proportional to the cell value within each panel. Panel (a): the sample is the co-authorship corpus of Appendix B.1. For each row, we take every paper with at least one author employed by one of the row segment’s three firms. Each cell then reports the percentage of those papers that also have an author employed by one of the column segment’s three firms. Reading along the EDA row, for example, 13.3% of papers with an EDA-firm author also have a co-author at one of the three foundries, and 9.2% have a co-author at one of the three equipment suppliers. Because each row uses its own segment’s papers as the denominator, the matrix is not symmetric: the same joint papers make up 13.3% of EDA-firm papers but 7.0% of foundry papers, since foundries publish more. Diagonal cells report papers co-authored by two distinct firms in the same segment. Blank cells indicate that no joint papers exist; the OSAT row is blank because the three OSAT firms publish no papers in the corpus. Panel (b): each cell counts disclosures of R&D collaboration identified in quarterly earnings-call transcripts (Appendix B.1) in which a firm in the row segment is the speaker and a firm in the column segment is the named partner. Counts are directional: the same relationship appears once for each party that discloses it. Because the measure requires a named partner and technical specificity, it is a lower bound on collaboration. Panel (c): each cell reports granted U.S. patents co-assigned to a row-segment and a column-segment firm, as a share of all granted patents held by the row segment’s firms. Data sources: OpenAlex; S&P Capital IQ via WRDS; PatentsView.

manufacturing process rather than firms competing for the same customer: all ten papers behind the equipment–equipment cell pair ASML, which supplies lithography systems, with either Applied Materials or Tokyo Electron, which supply deposition and etch equipment, and those two never appear on a paper together.

Finally, where the measures diverge, the divergence is in disclosure rather than in incidence. Disclosure in earnings calls is not symmetric between the parties to a collaboration: EDA firms disclose collaborations with foundries 46 times, foundries disclose collaborations with EDA firms three times — these are, for the most part, the same relationships seen from opposite sides. The party that discloses is the one for whom the relationship is a claim addressed to third parties—for an EDA firm, certification on a leading foundry’s process is the central claim it makes to the designers who buy its tools; for a foundry, access to a leading supplier’s equipment is a claim it makes to its own customers. Equipment suppliers show this point nicely: among the most connected firms in the co-authorship corpus, they disclose no collaborations of their own in earnings calls, while foundries name them seven times. What the earnings-call matrix measures is therefore jointly the incidence of collaboration and the incentive to disclose it. Its value is that it provides complementary evidence on where collaboration concentrates, and an independent signal that these collaborations carry economic value to the disclosing firms.

Taken together, these results show that semiconductor firms collaborate on R&D to a remarkably high degree and in exactly the way the technological interdependency described in Section 2 predicts. Collaboration is predominantly vertical; moreover, it is high where a firm cannot advance the technology frontier on its own. These findings establish a network of R&D collaboration through which knowledge *may* travel from one foundry to another. This point is most salient for EDA firms, a very concentrated market segment (with three dominant firms) that collaborates on R&D intensively with foundries. An EDA firm that works together with one foundry to bring a new technology node to market will create knowledge that may be useful for the next foundries it works with. High market concentration in the EDA segment implies the knowledge can spread through market mediated mechanisms to the many other foundries that use the same EDA supplier.

What we have not yet characterized, however, is how this knowledge is governed. Panel C shows that the results of these collaborations are not jointly patented. Co-assignment is uncommon in general, so this on its own would not be decisive. What makes it informative is that it is what the industry’s appropriability regime predicts. Semiconductor firms report patenting only 20.6 percent of their process innovations, and rate secrecy roughly two and a half times as effective as patents in protecting them; where they do patent, they do so over-

whelmingly to support cross-licensing and negotiation rather than to fence off a technology (Cohen et al., 2000).¹⁶

It is also not the case that publication in the engineering venues puts the co-developed knowledge in the public domain. Papers disclose the diagnostic and methodological layer of the joint work—measurement decompositions, optimization workflows, criteria for when a correction is worth deploying—while withholding the operating content required to reproduce it: results are reported in normalized units against undisclosed baselines, materials and product identities are anonymized, and process recipes are omitted, at times explicitly.¹⁷ What these papers attest is the work: who did what, at whose facility, and on which problem. What they withhold is what a reader would need in order to do it. This is consistent with survey evidence that semiconductor firms screen conference presentations and “refrain from publishing details that would permit exact duplication of the research results” (Appleyard, 1996, p. 138, n. 2).

The publication record and the patent record make a consistent point: the valuable output of these collaborations is neither disclosed nor jointly titled; rather, it is a business secret guarded closely by the firm, but importantly, available to its other customers if it chooses to. This is what makes the knowledge mobile. A patent excludes by law and discloses by design; secrecy does neither. It leaves the party holding the knowledge with discretion over who else obtains it—and in a vertical collaboration, that party is a supplier that sells to its collaborator’s competitors. What such a supplier carries onward is not its customer’s proprietary process, which contract restricts, but its own accumulated capability and the improvements embodied in the tools, models and design flows it sells to the rest of the industry.

4.2 The movement of engineers within and across firms

The second mechanism we turn to examines worker flows. The economics literature has established that knowledge is partly embedded in workers and travels with them when they change employers: hiring workers with experience at highly productive firms raises the productivity of the firms that hire them (Serafinelli, 2019), and the gains do not stop with the mover — when domestic firms hire workers trained at multinationals, the wages of incumbent workers rise as well, indicating that what the new hire brings diffuses inside the firm that hires her (Poole, 2013). The literature on the semiconductor industry reviewed in Section 2 documents the same channel in this industry, tracing it through the patents that

16. Figures for the semiconductor industry rest on 18 respondents (12–14 for patent propensity), and should be read accordingly.

17. Wong et al. (2013), an IBM–ASML collaboration, print the process flow used in their experiment with its fifth step listed as “Other process steps (not specified),” and report product results only as normalized values against an undisclosed baseline. ? report the performance of a newly developed mask-cleaning process without disclosing its composition. Ma et al. (2012) state design rules only in topological form, with no absolute dimension appearing in the paper.

mobile engineers file at their new employers (Almeida and Kogut, 1999; Song, Almeida, and Wu, 2003). That literature is itself divided on how to read these movements: Almeida and Kogut (1999) treat them as an externality of a fluid regional labor market, while Song et al. (2003) treat hiring as a deliberate strategy for sourcing knowledge.

What none of this work observes is the *timing* of hiring. This section takes up this question. Specifically, we examine whether firms deliberately hire engineering talent from abroad exactly when they need it — when they are pushing their firm-country technology frontier by upgrading their technology node, and, hence, have a deficit of accumulated process knowledge. The central feature of our research design is that we are able to observe when the deficit in accumulated process knowledge emerges using detailed, fab-level data that reports the minimum feature size (the technology node) operated by the fab. This data allows us to pinpoint when node-specific cumulative production experience for that firm-country pair is zero, as it is starting production in a new technology node. This timing helps distinguish between two channels: unintentional spillovers from a fluid labor market have no reason to synchronize with a firm’s technology upgrade, whereas sourcing process knowledge embedded in foreign workers through the market does.

Our purpose is two-fold. First, by tying labor flows to technology upgrading events, we explore a labor-flows explanation for the cross-border learning identified in the previous section. Second, the same exercise informs whether firms acquire expertise deliberately and through the market, or whether these knowledge flows are better understood as unintentional spillovers. This margin — hiring that responds to a firm’s needs — is treated in existing work as a threat to identification to be removed, not as an object of study (Poole, 2013; Serafinelli, 2019).

Our empirical analysis examines how *cross-border inflows* of semiconductor engineers respond when a firm–country pair (e.g., TSMC, USA) introduces a technology node more advanced than it has previously produced. We exclude cases where the technology leader, TSMC Taiwan, is the treated unit.

We identify technology upgrades using SEMI’s World Fab Watch, a proprietary census of front-end semiconductor manufacturing facilities that reports, at the fab-quarter level, technology node, capacity, and product type. We aggregate these data to the firm–country–year level and define an upgrade as a decline in the smallest feature size operated by a firm in a country relative to the previous year.

We measure labor flows using Revelio Labs’ individual employment histories, accessed through Wharton Research Data Services. Revelio provides the employer, establishment location, start and end dates, occupational role, and seniority for each position. We use all positions at firms classified as semiconductor and related device manufacturers (NAICS 334413), covering 6,092 firms and 3.77 million position records. An arrival is classified as

cross-border when the worker’s previous reported position was in a different country, where country refers to the location of the establishment rather than the employer’s headquarters.

We construct four measures of cross-border inflows intended to capture workers likely to possess process-engineering knowledge relevant to an upgrading fab. All measures restrict attention to workers with seniority (excluding interns and entry-level positions). The first (*Senior Engineer*) includes all arrivals whose occupation contains “engineer”. The second (*Senior Engineer SEMI*) further requires that the worker’s previous employer operate a fab in the SEMI data. The third (*Senior Core and Related SEMI*) additionally restricts occupations to role clusters related to semiconductor manufacturing, including occupations both directly involved in fab operations and adjacent technical roles. The fourth and most restrictive (*Senior Core SEMI*), retains only the roles directly tied to fab operations.

Matching firm–country pairs across SEMI and Revelio yields a firm–country–year panel of cross-border labor inflows containing 236 technology-upgrade events over 2004–2024. Further details on the data and variable construction are provided in Appendix B.2.

Research design and estimation results

To understand how cross-border labor flows respond to technology upgrading events, we compare foreign worker arrivals at upgrading and non-upgrading firm-country pairs. In our setting, treatment is non-absorbing: a firm–country pair can upgrade technology multiple times. Thus, we follow Cengiz, Dube, Lindner, and Zipperer (2019) and estimate stacked event study specifications that allow for non-absorbing treatment. We create a separate panel dataset for every valid technology upgrading event, paired with all clean controls in a predefined time window around the event. These event specific datasets are then “stacked” to create the panel for estimation.

In our setting, the window captures six years in total: three years before treatment, the year of the upgrading event, and two years following treatment. A valid technology upgrading event is one that has a clean pre-treatment period, meaning no previous technology upgrades in the preceding three years.¹⁸ Clean firm-country controls are those without an upgrading event in the full six-year window.

We estimate specifications of the following form

$$L_{fctd} = \alpha + \sum_{k \in \{-3, -2, 0, 1, 2\}} \beta_k (D_{td}^k \times Treated_{fcd}) + \gamma_{fcd} + \delta_{td} + \epsilon_{fctd} \quad (10)$$

18. Following the literature (e.g., Blau, Cohen, Comey, Kahn, and Boboshko (2026)), we allow for further upgrades in the post-period, which affects the interpretation of our coefficients of interest, capturing the combined effect of the upgrades.

where fc indexes the firm-country pair, t indexes the year and d indexes the event-specific dataset. The dependent variable L_{fcd} is the count of the cross-border inflow of workers to firm-country pair fc in year t , as it enters dataset d . The indicator D_{td}^k equals one when year t lies k years from the upgrading event that defines dataset d , and $Treated_{fcd}$ equals one for the pair that upgrades in dataset d and zero for that dataset's clean controls. We omit $k = -1$, so each β_k measures the difference in inflows between upgrading pairs and their clean controls relative to the year before the event, averaged across datasets. The firm-country and year fixed effects are both interacted with the event-specific dataset so that only within dataset d variation is used for estimation. Standard errors are clustered at the firm-country pair by event level.

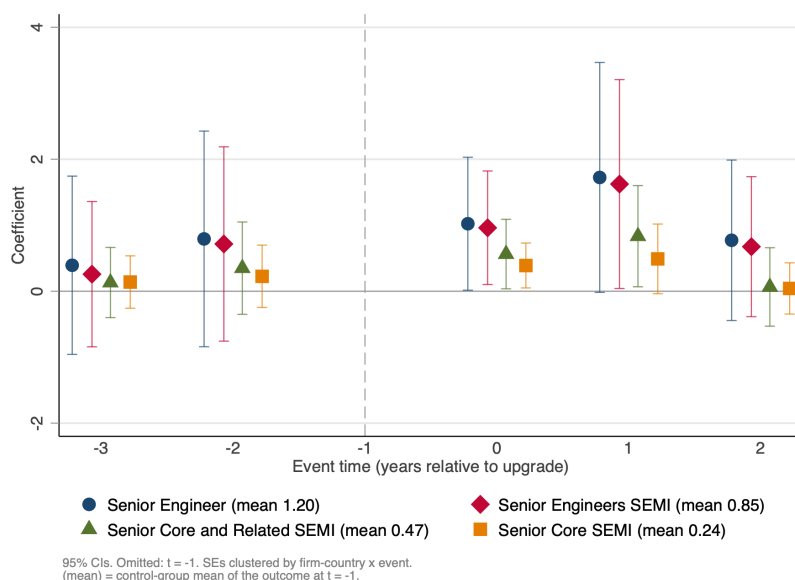


Figure 5: Cross-border skilled, senior labor flows around technology upgrading events

Figure 5 shows the baseline estimates. All four measures have flat pretrends. Before the upgrade, treated and non-treated firms display similar cross-border hiring patterns lending support to our research design.¹⁹ All four measures show that cross-border inflows of engineers are elevated for upgrading firms for two years: in the year of the upgrade, as well as the subsequent year. By the second year following the upgrade, there is no differential effect on hiring. All point estimates fall towards zero, and none are statistically significant, providing some validation for our choice of the size of the treatment windows.

19. In theory, anticipation effects are possible: firms may hire skilled workers before entering volume production, yet there is a mechanism pushing in the opposite direction. In particular, the fact that pretrends are flat at the annual level may reflect the lag between when a worker is hired and when they start working (and show up in our data). For foreign workers in particular, this time lag can be both lengthy and uncertain due to immigration and visa processing times.

Importantly, the different measures of skilled labor respond in a manner consistent with the deliberate sourcing of cross-border knowledge through workers. The broadest, and arguably noisiest measure of workers with relevant process knowledge are senior engineers. Our estimates show that an upgrading firm adds about one additional senior engineer from abroad in the year of the upgrade, and two in the following year (though the effects are noisily estimated). Relative to a mean of 1.2 in the control group in the year preceding the upgrade, that's an economically meaningful effect. Using the labor flow measures that more narrowly target relevant fab experience yields larger effects relative to control group means. For example, based on the narrowest measure of fab-relevant occupations, upgrading fabs add, on average, about half an additional worker in both years, relative to a mean of 0.24 in the control group in the year preceding the upgrade. Thus, while these measures capture a thinner slice of labor flows, they tend to respond more strongly relative to their size. This suggests that when upgrading, firms target their cross-border hiring to workers with previous experience in fab operations. Our baseline results pool memory and logic producing firms. In Appendix Figure B.2, we show results that restrict to logic upgrades and find similar effects using a much smaller set of upgrading events: elevated hiring for two years, with estimates falling towards zero in final year of the estimation window.²⁰

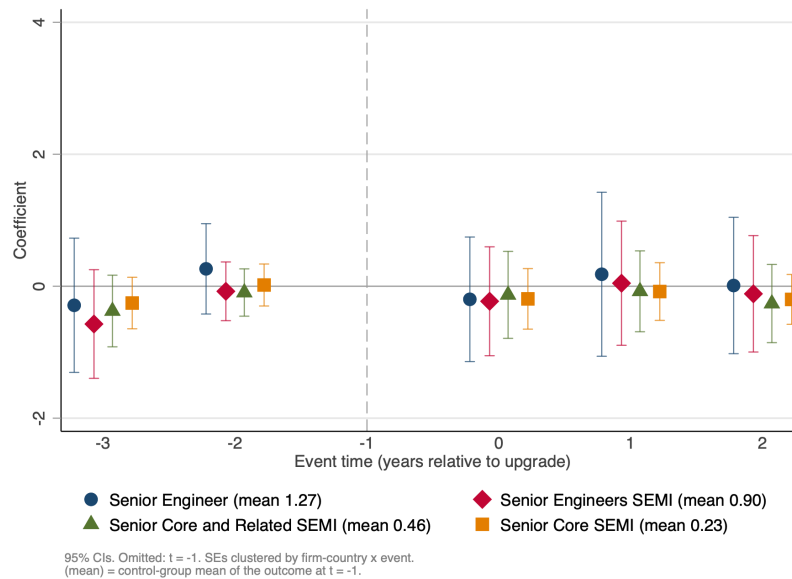


Figure 6: Placebo: Cross-border skilled, senior labor flows during capacity expansion without technology upgrading

A competing explanation for our findings is that we are picking up elevated hiring when a firm expands production. That is, as technology upgrading events also expand the production capacity of firms, we cannot disentangle which of these two mechanisms drives our results in 20. Given the much smaller sample — 46 events as opposed to 236, the estimates are correspondingly noisier.

the baseline estimates. To investigate this possibility, we examine placebo events that expand a firm-country pair's total capacity *without* technology upgrading. Specifically, we define treatment as the entry of a new fab within the firm-country's technology frontier. Figure 6 shows that capacity expansion events do not result in differential cross-border hiring for any of our measures. Coefficients cluster around zero, with no evident trend, and none of the coefficients are statistically different from zero. Taken together with the baseline results, this shows that firms disproportionately hire senior skilled workers from abroad when they upgrade their technology, not when they expand production capacity.

Finally, we use the granular data on positions from Revelio to distinguish two margins of cross-border hiring: internal moves between multinational affiliates and external hires. Figure 7 plots the estimated coefficients for senior engineers. The effect for within firm and across firm hiring show a similar pattern with elevated hiring concentrated in the year of the upgrade and the following year (though the confidence intervals cross zero for internal moves). Together, the results suggest complementary channels of knowledge flows. On the one hand, the precisely estimated increase in the hiring of external labor points to firms deliberately poaching talent from rival firms precisely when they lack accumulated process knowledge. On the other hand, though imprecisely estimated, our results also point to the important role of multinational firms in mediating cross-border knowledge flows. This suggests that there may be tacit, uncodifiable aspects of process knowledge: a multinational firm could simply transmit its blueprints to the upgrading affiliate. The fact that our results suggest they also send skilled, senior engineers is consistent with process knowledge embedded in workers. In the next and final section, we examine more closely the role of voluntary and deliberate technology sharing by firms across borders.

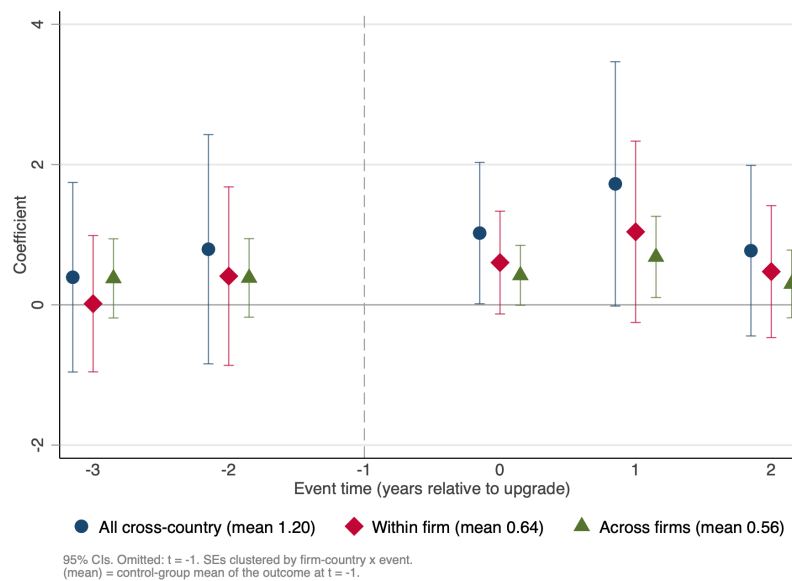


Figure 7: Internal and external hiring margins

4.3 Deliberate Technology Sharing Across Borders by Firms

The previous sections documented workers collaborating on R&D across countries within a single multinational firm as well as multinationals sharing process knowledge across borders embedded in skilled workers. These are but two among many well-documented examples of firms voluntarily and deliberately sharing knowledge across borders in the industry.

The most striking evidence pointing to the importance of firms sharing technology across borders comes from the historical record.²¹ With the important exception of the United States, where the industry originated, the countries that subsequently developed major semiconductor industries did not build their technological capabilities in isolation. Their trajectories differed substantially, as did the role played by their governments, but Japan, Korea, and Taiwan all relied heavily on deliberate acquisition and absorption of technology developed abroad. Meeting the demand for foreign technology from follower countries relied, in turn, on US firms at the technology frontier being willing to share that technology, and policymakers allowing (or pressuring) them to do so.

In Japan the government facilitated access to technological knowledge that domestic firms did not initially possess with MITI (the Ministry of International Trade and Industry) directly administering imports of semiconductor technology during the industry's early development. Japanese firms licensed technology from foreign producers, with the government helping to coordinate agreements and guarantee payments (Lynn, 1998). MITI also intervened in the terms of many licensing agreements, strengthening the bargaining position of Japanese firms (Lynn, 1998; Choi, 2008).

Korea similarly combined domestic support with intensive international technology acquisition in its efforts to reach the technology frontier. The government made semiconductors a strategic priority and established the Electronics and Telecommunication Research Institute (ETRI) as part of its broader effort to move beyond assembly (Amsden, 1989). Korean producers formed numerous technological alliances with foreign firms and acquired firms in the United States in order to obtain access to advanced technologies and experienced scientists and engineers (Chen and Sewell, 1996). In this case, domestic capability building involved both the accumulation of Korean production experience and deliberate links to the existing international technological frontier.

The importance of foreign technology is perhaps clearest in Taiwan. The government established a public pilot facility through what later became the Electronics Research and Service Organization (ERSO), which developed semiconductor manufacturing capabilities in cooperation with RCA in the United States (Liu, 1993). The technological capabilities accumulated through this relationship were subsequently transferred to domestic firms through licensing and spin-offs, including UMC and TSMC (Liu, 1993). Taiwan's eventual emer-

21. For a detailed discussion of the historical evidence, see Goldberg et al. (2024).

gence at the technological frontier was thus built on a deliberate institutional mechanism for obtaining foreign technology, learning how to use it, and transferring the resulting capabilities to domestic producers.

These experiences demonstrate that international knowledge transmission is not automatic. Access depended on foreign firms and governments being willing to license technology, form alliances, train workers, invest, or otherwise participate in technology transfer. U.S. antitrust policy helped make technology available to foreign firms (Choi, 2008), while Japanese firms were subsequently more reluctant to transfer technology to emerging Korean competitors in anticipation of future competition (Chen and Sewell, 1996).

The experience of China reinforces this point from the opposite direction. China has provided extensive support to its domestic semiconductor industry for decades and has repeatedly sought partnerships and joint ventures with foreign firms in an effort to obtain foreign technology (Minnich, 2023). Yet its ability to obtain frontier technology has historically been more limited (Miller, 2022; Minnich, 2023). The contrast is informative: generous domestic support may stimulate production and hence domestic learning, but it need not substitute for access to technological capabilities accumulated elsewhere.

Taken together, the historical evidence points to a striking conclusion: outside the United States, it does not identify a major semiconductor producer that reached the technological frontier without substantial access to foreign technology. This finding also reinforces the main insight conveyed by the counterfactual experiment of restricting China's access to cross-border learning in advanced technologies (Section 3.6.2), and suggests that efforts to do so may prove highly effective.

Contemporary evidence points to a similar trend playing out in the US currently. Intended in part to address the reversal of fortune in advanced manufacturing capabilities between the US and Taiwan, the United States authorized a semiconductor manufacturing programme in the CHIPS for America Act of January 2021 and funded it in the CHIPS and Science Act of 2022. Informed by the historical narrative, we ask whether this most recent effort to acquire advanced manufacturing capabilities by a technological follower country has resulted in increased cross-border technology sharing by firms.

To do so, we use data on greenfield FDI events from *fDi Markets* which records announcements of FDI. This measure is intended to capture two forms of knowledge sharing by multinational firms. First, and most directly, knowledge may be shared cross-border by establishing advanced node fabs in the US. When an MNC such as TSMC establishes US-based fabs at a more advanced technology node than that currently operated in the US, it is directly advancing the US technology frontier. Second, knowledge from other parts of the value chain may also be shared through FDI.

Figure 8 Panel A plots the count of inbound FDI in the semiconductor industry for the US and other countries 2004-2024. There is a clear surge in greenfield FDI that ramps up after

2021 in the US that is not comparable to that observed in other countries.^{22,23} There were, on average, 6 FDI events annually up to 2020, and 13.25 after pointing to a more than doubling of FDI in the sector: policies intended to advance production capabilities in the US coincided with a surge in inbound FDI.

Importantly, the inflow of FDI spans the value chain. Alongside FDI in frontier fabrication, most notably TSMC, we observe investment by silicon wafer producers (GlobalWafers), capital equipment and optics suppliers (ASML, ASM International, Tokyo Electron, Carl Zeiss), and by imec, which established an R&D facility in Michigan (Panel B plots the count of inbound FDI for Semiconductor Machinery and shows a similar trend, though the surge is concentrated in one year). This suggests that countries behind the technology frontier cannot acquire frontier capability by importing equipment and inputs alone: the firms that supply them and the institutions that organize collaboration among them relocate as well. This is consistent with the closer-than-arm's-length collaboration documented in Section 4.1, where suppliers' engineers work inside the customer's fab.

Finally, this evidence also speaks to the counterfactual in Section 3.6.2 that examined the effects of supporting domestic manufacturing. Consistent with an important role for cross-border knowledge flows, our finding on elevated FDI in the US shows that, rather than relying on domestic production experience to increase process knowledge, current US efforts rely on foreign process knowledge partly mediated through multinational firms.

5 Conclusions

Learning-by-doing has long been recognized as a defining feature of semiconductor manufacturing. Yet discussions of learning in this industry have generally emphasized knowledge accumulation within firms or spillovers among firms located in the same country. In this paper, we argue that this perspective is incomplete. Modern semiconductor manufacturing is fundamentally embedded in an international production system in which technological capabilities are accumulated not only through domestic experience, but also through extensive cross-border learning.

We establish this conclusion using three complementary approaches. We first review industry evidence along with the historical development of the semiconductor industry and show that, with the exception of the United States, every major producer built its manufacturing capabilities through sustained access to frontier technology developed abroad. We

22. The one other country showing a similar, though smaller uptick is Germany, which also adopted policies aimed to increase semiconductor manufacturing capabilities.

23. We also estimated synthetic control and synthetic difference-in-differences specifications, reported in Appendix Figure B.3. However, we do not rely on them as the pre-treatment fit is poor.

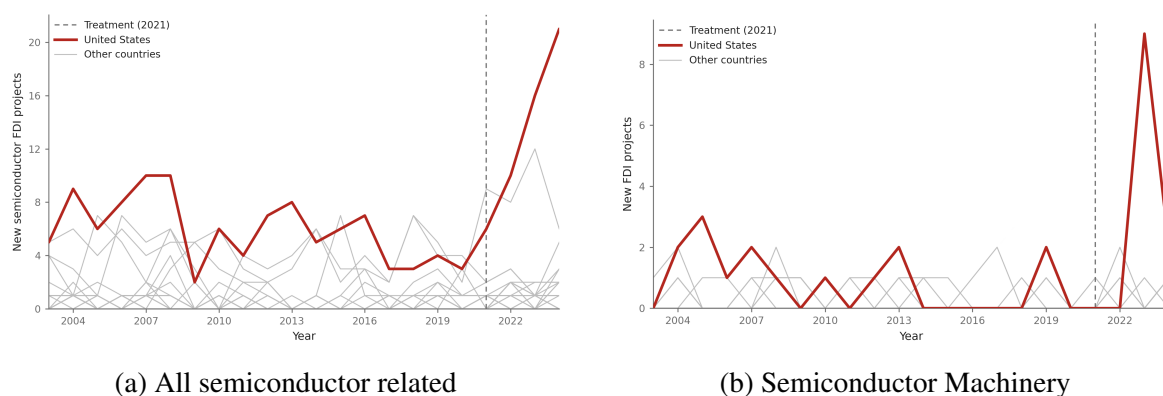


Figure 8: Semiconductor greenfield FDI projects

Notes: *Panel A* documents the annual count of announced greenfield FDI projects by destination country in all semiconductor-related activities, from fDi Markets. Grey lines are 26 comparison countries: Australia, Austria, Belgium, Bulgaria, Canada, Czechia, Estonia, Finland, France, Germany, Ireland, Italy, Latvia, Morocco, Netherlands, Norway, Poland, Romania, Slovakia, Slovenia, South Africa, Spain, Sweden, Switzerland, Thailand and the United Kingdom. The vertical line marks the 2021 Chips for America Act. *Panel B* is the same figure but restricts to projects that involve the production of semiconductor machinery.

then estimate a stylized empirical model that quantifies the contribution of cross-border learning relative to own-technology learning and learning across technologies within the supplier. Finally, we provide reduced-form evidence on the principal mechanisms through which international knowledge transmission occurs. Despite relying on different sources of evidence and distinct empirical approaches, all three analyses point to the same conclusion: cross-border learning is an important feature of modern semiconductor manufacturing.

A particularly striking result is that several important channels of cross-border learning appear to be market-mediated. Our evidence points to firms' organizational and investment decisions – including collaborative innovation along global supply chains, the international mobility of engineers, often within multinational enterprises, and foreign direct investment – as potentially important mechanisms of knowledge transmission. These channels differ from the traditional view of Marshallian knowledge spillovers, according to which knowledge is "in the air" and diffuses passively through geographic proximity without deliberate action or compensation. Rather than being highly localized and largely accidental, an important part of the relevant knowledge transmission in semiconductor manufacturing reflects purposeful economic relationships that frequently span national borders.

This perspective has broader implications for how knowledge diffusion is understood in globally integrated industries. International production networks do more than allocate production across countries; they also facilitate the transmission of technological capabilities. Firms actively invest in global networks because they internalize the private returns from sharing knowledge across facilities, suppliers, and workers. In this sense, global value chains

are not simply organizational structures for coordinating production but also institutions through which technological progress is generated and disseminated internationally.

These findings also have important implications for current policy debates. To the extent that knowledge transmission is internalized through market relationships, the conventional justification for industrial policy based on non-internalized technological spillovers becomes less compelling. At the same time, governments do not internalize the benefits that domestic technological advances generate abroad. National governments evaluate industrial policies through the lens of domestic objectives, while firms organize production and knowledge flows globally. This divergence creates a natural tension between the incentives of globally integrated firms and the objectives of governments pursuing resilience, technological leadership, or national security.

The existence of international knowledge spillovers also implies that the consequences of industrial policy depend critically on the perspective from which they are evaluated. From the standpoint of an individual country, the international diffusion of knowledge may weaken the attractiveness of policies whose benefits partially accrue to geopolitical competitors. From the perspective of multilateral institutions concerned with global welfare, however, the same spillovers imply positive international externalities. Policies that successfully foster technological progress in one country may accelerate capability building elsewhere. Taiwan's long-standing support for the development of TSMC, for example, helped establish an industrial model whose benefits ultimately extended well beyond Taiwan through the international diffusion of manufacturing capabilities.

Finally, because important channels of cross-border learning operate through market mechanisms rather than anonymous spillovers, it can be deliberately promoted or deliberately disrupted. This observation is particularly relevant in the current policy environment. Measures that restrict access to international production networks, such as export controls or other forms of weaponized interdependence, have the potential to substantially slow technological catch-up by limiting access to frontier knowledge and advanced manufacturing capabilities. Conversely, policies that attract or expand production by firms already operating at the technological frontier may be especially effective because these firms bring with them access to accumulated knowledge, established organizational capabilities, and internal channels of cross-border learning. In semiconductor manufacturing, therefore, policies affect not only where production takes place, but also how—and whether—knowledge itself continues to flow across borders.

Overall, our results suggest that understanding semiconductor manufacturing requires shifting attention from the geography of production to the geography of knowledge transmission. In an industry where technological capabilities increasingly depend on international production networks, policies that reshape those networks will inevitably reshape the global distribution of technological progress. This insight extends beyond semiconductors and

may prove useful for thinking about knowledge diffusion in other technologically advanced industries that rely on globally integrated production systems.

References

- Aitken, B. J. and A. E. Harrison (1999). Do domestic firms benefit from direct foreign investment? evidence from venezuela. *American Economic Review* 89(3), 605–618.
- Alfaro, L. and M. X. Chen (2018). Selection and market reallocation: Productivity gains from multinational production. *American Economic Journal: Economic Policy* 10(2), 1–38.
- Alfaro, L., H. Fadinger, J. S. Schymik, and G. Virananda (2025, May). Trade and industrial policy in supply chains: Directed technological change in rare earths. NBER Working Paper 33877, National Bureau of Economic Research. Revised March 2026.
- Almeida, P. and B. Kogut (1999). Localization of knowledge and the mobility of engineers in regional networks. *Management Science* 45(7), 905–917.
- Amsden, A. H. (1989). *Asia's next giant: South Korea and late industrialization*. Oxford University Press, USA.
- Antràs, P. (2021). De-globalisation? global value chains in the post-covid-19 age. *Annual Review of Economics* 13, 427–459.
- Antràs, P. and D. Chor (2013). Organizing the global value chain. *Econometrica* 81(6), 2127–2204.
- Appleyard, M. M. (1996). How does knowledge flow? interfirm patterns in the semiconductor industry. *Strategic Management Journal* 17(Winter Special Issue), 137–154.
- Appleyard, M. M. (2003). The influence of knowledge accumulation on buyer-supplier codevelopment projects. *Journal of Product Innovation Management* 20(5), 356–373.
- Arrow, K. J. (1962). The economic implications of learning by doing. *Review of Economic Studies* 29(3), 155–173.
- Barwick, P. J., M. Kalouptsidi, and N. B. Zahur (2025). Industrial policy implementation: Empirical evidence from china's shipbuilding industry. *The Review of Economic Studies* 92(6), 3611–3648.
- Beaumier, G. and M. Cartwright (2024). Cross-network weaponization in the semiconductor supply chain. *International Studies Quarterly* 68(1).
- Benkard, C. L. (2000). Learning and forgetting: The dynamics of aircraft production. *American Economic Review* 90(4), 1034–1054.
- Besanko, D., U. Doraszelski, and Y. Kryukov (2019). How efficient is dynamic competition? the case of price as investment. *American Economic Review* 109(9), 3339–3364.
- Besanko, D., U. Doraszelski, Y. Kryukov, and M. Satterthwaite (2010). Learning-by-doing, organizational forgetting, and industry dynamics. *Econometrica* 78(2), 453–508.
- Blau, F. D., I. Cohen, M. Comey, L. M. Kahn, and N. Boboshko (2026). The minimum wage and inequality between groups. *ILR Review* 79(5), 756–790.
- Bloom, N., M. Schankerman, and J. Van Reenen (2013). Identifying technology spillovers and product market rivalry. *Econometrica* 81(4), 1347–1393.
- Bonvillian, W. B. (2024). The u.s. chips and science act: What is it trying to do? *Issues in Science and Technology* 40(2).
- Bonvillian, W. B. and S. Sarma (2023). *The Chip Design Revolution*. Cambridge, MA: MIT Press.

- Bown, C. P. (2020). How the United States marched the semiconductor industry into its trade war with China. *East Asian Economic Review* 24(4), 349–388.
- Cengiz, D., A. Dube, A. Lindner, and B. Zipperer (2019). The effect of minimum wages on low-wage jobs. *The quarterly journal of economics* 134(3), 1405–1454.
- Chen, C.-F. and G. Sewell (1996). Strategies for technological development in south korea and taiwan: the case of semiconductors. *Research Policy* 25(5), 759–783.
- Choi, H. (2008). Technology Importation, Corporate Strategies, and the Rise of the Japanese Semiconductor Industry in the 1950s. *Comparative Technology Transfer and Society* 6(2), 103–126.
- Clayton, C., M. Maggiori, and J. Schreger (2026). A framework for geoeconomics. *Econometrica* 94(1), 105–136.
- Coe, D. T. and E. Helpman (1995). International r&d spillovers. *European Economic Review* 39(5), 859–887.
- Cohen, W. M., R. R. Nelson, and J. P. Walsh (2000, February). Protecting their intellectual assets: Appropriability conditions and why U.S. manufacturing firms patent (or not). NBER Working Paper 7552, National Bureau of Economic Research.
- Collard-Wexler, A., G. Gowrisankaran, and R. S. Lee (2019). Nash-in-nash bargaining: A microfoundation for applied work. *Journal of Political Economy* 127(1), 163–195.
- Deng, S., D. Jia, M. Leccese, and A. Sweeting (2024, April). Bargaining and dynamic competition. Working Paper 32360, National Bureau of Economic Research.
- Fajgelbaum, P., P. Goldberg, P. Kennedy, A. Khandelwal, and D. Taglioni (2024, June). The us-china trade war and global reallocations. *American Economic Review: Insights* 6(2), 295–312.
- Fajgelbaum, P. D., P. K. Goldberg, P. J. Kennedy, and A. K. Khandelwal (2020, 02). The return to protectionism*. *The Quarterly Journal of Economics* 135(1), 1–55.
- Farrell, H. and A. L. Newman (2019a, 07). Weaponized interdependence: How global economic networks shape state coercion. *International Security* 44(1), 42–79.
- Farrell, H. and A. L. Newman (2019b). Weaponized interdependence: How global economic networks shape state coercion. *International Security* 44(1), 42–79.
- Goldberg, P. K., R. Juhász, N. J. Lane, G. Lo Forte, and J. Thurk (2024). Industrial policy in the global semiconductor sector. NBER Working Paper 32651, National Bureau of Economic Research. Revised February 2026.
- Government of the Netherlands (2023, March). Letter to parliament on additional export control measures concerning advanced semiconductor manufacturing equipment.
- Guldi, R. L. (2004, November). In-line defect reduction from a historical perspective and its implications for future integrated circuit manufacturing. *IEEE Transactions on Semiconductor Manufacturing* 17(4), 629–640.
- Hallak, J. C. and P. K. Schott (2011). Estimating cross-country differences in product quality. *Quarterly Journal of Economics* 126, 417–474.
- Hughes, T. P. (1983). *Networks of Power: Electrification in Western Society, 1880–1930*. Baltimore: Johns Hopkins University Press.
- Hummels, D. and P. J. Klenow (2005). The variety and quality of a nation’s exports. *American Economic Review* 95, 704–723.
- Irwin, D. A. and P. J. Klenow (1994). Learning-by-doing spillovers in the semiconductor industry. *Journal of Political Economy* 102(6), 1200–1227.

- Javorcik, B. S. (2004). Does foreign direct investment increase the productivity of domestic firms? in search of spillovers through backward linkages. *American Economic Review* 94(3), 605–627.
- Johnson, R. (2012). Trade and prices with heterogeneous firms. *Journal of International Economics* 86(1), 43–56.
- Juhász, R., N. Lane, and D. Rodrik (2024). The new economics of industrial policy. *Annual Review of Economics* 16, 213–242.
- Kalouptsi, M. (2018). Detection and impact of industrial subsidies: The case of chinese shipbuilding. *The Review of Economic Studies* 85(2), 1111–1158.
- Kapoor, R. and P. J. McGrath (2014). Unmasking the interplay between technology evolution and R&D collaboration: Evidence from the global semiconductor manufacturing industry, 1990–2010. *Research Policy* 43, 555–569.
- Keller, W. (2002). Geographic localization of international technology diffusion. *American Economic Review* 92(1), 120–142.
- Keller, W. (2004). International technology diffusion. *Journal of Economic Literature* 42(3), 752–782.
- Khandelwal, A. (2010). The long and short (of) quality ladders. *Review of Economic Studies* 77, 1450–1476.
- Kim, J.-S., W.-K. Ma, Y.-S. Kim, M.-S. Kim, W.-T. Kwon, S.-K. Park, P. Nikolsky, M. Otter, M. Escalante Marun, R. Anunciado, K.-T. Sun, G. Storms, and E. van West (2014). CD uniformity optimization at volume ramp up stage for new product introduction. In *Metrology, Inspection, and Process Control for Microlithography XXVIII*, Volume 9050 of *Proc. SPIE*, pp. 905037.
- Kugler, M. and E. Verhoogen (2011). Prices, plant size, and product quality. *Review of Economic Studies* 79, 307–339.
- Liebmann, L. W., J. Kye, B.-S. Kim, L. Yuan, and J.-P. Geronimi (2010). Taming the final frontier of optical lithography: Design for sub-resolution patterning. In *Design for Manufacturability through Design-Process Integration IV, Proceedings of SPIE*, Volume 7641, pp. 764105.
- Liu, C.-Y. (1993). Government's role in developing a high-tech industry: the case of taiwan's semiconductor industry. *Technovation* 13(5), 299–309.
- Liu, J., M. Rotenberg, and S. Traiberman (2026). Sabotage as industrial policy. Working paper.
- Lynn, L. H. (1998). Japan's technology-import policies in the 1950s and 1960s: did they increase industrial competitiveness? *International Journal of Technology Management* 15(6-7), 556–567.
- Ma, Y., J. Sweis, H. Yoshida, Y. Wang, J. Kye, and H. J. Levinson (2012). Self-aligned double patterning (SADP) compliant design flow. In *Design for Manufacturability through Design-Process Integration VI, Proceedings of SPIE*, Volume 8327, pp. 832706.
- Markusen, J. R. (2002). *Multinational Firms and the Theory of International Trade*. Cambridge, MA: MIT Press.
- Miller, C. (2022). *Chip War: The Fight for the World's Most Critical Technology*. Scribner.
- Minnich, J. D. (2023). *Re-Innovation Nation: The Strategic and Political Logic of Technology Transfer Policy in Rising China*. Ph. D. thesis, Massachusetts Institute of Technology.
- Miravete, E. J., M. J. Moral, and J. Thurk (2018). Fuel taxation, emissions policy, and competitive advantage in the diffusion of european diesel automobiles. *The RAND Journal of Economics* 49, 504–540.

- OECD (2019). Measuring distortions in international markets: The semiconductor value chain. Technical Report 234, OECD Publishing, Paris.
- OECD (2019). Measuring distortions in international markets: The semiconductor value chain. (234).
- Oettl, A. and A. Agrawal (2008). International labor mobility and knowledge flow externalities. *Journal of International Business Studies* 39(8), 1242–1260.
- Pillai, U. (2023). The origins of the tools suppliers in the semiconductor industry. *Business History* 65(6), 959–982.
- Poole, J. P. (2013, 05). Knowledge transfers from multinational to domestic firms: Evidence from worker mobility. *The Review of Economics and Statistics* 95(2), 393–406.
- Rosenberg, N. (1963). Technological change in the machine tool industry, 1840–1910. *The Journal of Economic History* 23(4), 414–443.
- Serafinelli, M. (2019). “good” firms, worker flows, and local productivity. *Journal of Labor Economics* 37(3), 747–792. No DOI printed in the published PDF; fields taken from the printed opening page and the running head of the final page.
- Song, J., P. Almeida, and G. Wu (2003). Learning-by-hiring: When is mobility more likely to facilitate interfirm knowledge transfer? *Management Science* 49(4), 351–365.
- Sullivan, J. (2023, April). Remarks by national security advisor jake sullivan on renewing american economic leadership at the brookings institution. Remarks at the Brookings Institution.
- Sweeting, A., D. Jia, S. Hui, and X. Yao (2022, April). Dynamic price competition, learning-by-doing, and strategic buyers. *American Economic Review* 112(4), 1311–33.
- Thurk, J. (2024). Outsourcing, firm innovation, and industry dynamics in the production of semiconductors. *Working Paper*.
- U.S. Department of Commerce, Bureau of Industry and Security (2022, October). Commerce implements new export controls on advanced computing and semiconductor manufacturing items to the people’s republic of china (PRC).
- U.S. Department of the Treasury (2024, October). Treasury issues regulations to implement executive order addressing U.S. investments in certain national security technologies and products in countries of concern.
- Varas, A., R. Varadarajan, J. Goodrich, and F. Yinug (2021, April). Strengthening the global semiconductor supply chain in an uncertain era. Technical report, Boston Consulting Group and Semiconductor Industry Association. Retrieved from semiconductors.org, 2026-07-30.
- Weber, C. (2004, November). Yield learning and the sources of profitability in semiconductor manufacturing and process development. *IEEE Transactions on Semiconductor Manufacturing* 17(4), 590–596.
- Wong, C., G. Seevaratnam, T. Wiltshire, N. Felix, T. Brunner, P. Rawat, M. Escalante, A. Elmalk, V. Wang, C. Leewis, P. Hinnen, W. Kim, and E. Rottenkolber (2013). CD optimization methodology for extending optical lithography. In *Metrology, Inspection, and Process Control for Microlithography XXVII*, Volume 8681 of *Proc. SPIE*, pp. 868137.
- World Bank (2020). *World Development Report 2020: Trading for Development in the Age of Global Value Chains*. Washington, DC: World Bank.

A Additional Model Estimation Details and Results

A.1 Construction of “Quality Index”

To construct the quality index $\bar{z}$ at the supplier-technology-quarter level, we make use of the *transactions-level* data. As noted in the data section, each observation in our data refers to a specific transaction, in which the buyer is anonymized; in the implementation of the empirical model, we aggregate the data to the product level jkt , but we use the transactions-level data when we construct our quality index. Let i denote the (unknown to us) buyer in a specific transaction “ $ijkt$ ” involving the purchase of chips of technology k from supplier j by buyer i at time t , P_{ijkt} be the transactions-specific price, and Q_{ijkt} be the transactions-specific quantity of chips. The quality index for product jkt is constructed using three characteristics, number of masks, number of metal layers, and number of polysilicon layers, in the following five steps:

1. Residual variation of metal layers and polysilicon layers

Because the number of masks ($\tilde{m}_{ijkt}$), number of metal layers ($metal_{ijkt}$), and number of polysilicon layers ($poly_{ijkt}$) are highly correlated, we first regress each of $metal_{ijkt}$ and $poly_{ijkt}$ on $\tilde{m}_{ijkt}$ to obtain the residual variation:

$$metal_{ijkt}, poly_{ijkt} \Rightarrow \hat{u}_{ijkt}^L, \hat{u}_{ijkt}^P$$

2. “Hedonic” Price Regression

$$\log P_{ijkt} = \beta_P \log Q_{ijkt} + \gamma_m \tilde{m}_{ijkt} + \gamma_L \hat{u}_{ijkt}^L + \gamma_P \hat{u}_{ijkt}^P + \text{FE} + e_{ijkt}$$

FE includes time fixed effects and supplier-technology fixed effect interactions.

We control for the quantity Q_{ijkt} of each transaction to account for the impact of potential quantity-based discounts on prices.

3. Transaction-level quality

$$\bar{z}_{ijkt} = \hat{\gamma}_m \tilde{m}_{ijkt} + \hat{\gamma}_L \hat{u}_{ijkt}^L + \hat{\gamma}_P \hat{u}_{ijkt}^P$$

4. Product-level Quality

$$\bar{z}_{jkt} = \sum_{i \in j} \log(Q_{ijkt}/q_{jkt}) \cdot \bar{z}_{ijkt}$$

5. We convert the log-quality estimate constructed in Step 4 into a strictly positive quality index, normalized so that the median quality in the first period equals 100.

A.2 Estimation Algorithm

We solve the model as follows conditional on a guess of the learning-by-doing parameters $\{\gamma, \alpha, \mu\}$.

1. Estimate $\{\hat{\sigma}, \hat{\delta}, \hat{\nu}_j^D\}$ via Equation 1 using 2SLS. The instruments used are described under “Estimation and Identification” in the main text. This estimation is unbiased provided unobserved quality $\Delta\xi_{jkt}$ is not serially correlated. As buyers maximize static utility, these demand estimates do not depend on $\{\gamma, \alpha, \mu\}$.
2. We recover $\{\hat{\eta}\}$ along with the technology fixed effects in Equation 2 as follows:
 - (a) Recover $\hat{\xi}$ from estimates of Equation 1; i.e., $\hat{\xi}_{jkt} = \hat{z}_{jkt}^{\hat{\delta}} \times \hat{\nu}_j^D \times \hat{\Delta}\xi_{jkt}$
 - (b) Construct the technology-level price index $\hat{P}_{kt} = \left[\sum_{j \in k} \left(\frac{p_{jkt}}{\hat{\xi}_{jkt}} \right)^{1-\hat{\sigma}} \right]^{\frac{1}{1-\hat{\sigma}}}$
 - (c) Construct the technology-level quantity index $\hat{Q}_{kt} = \left(\sum_{j \in k} (\hat{\xi}_{jkt} q_{jkt})^{\frac{\hat{\sigma}-1}{\hat{\sigma}}} \right)^{\frac{\hat{\sigma}}{\hat{\sigma}-1}}$
 - (d) Estimate $\{\hat{\eta}, \hat{\phi}_k^D\}$ via Equation 2 using 2SLS. The instruments used are described under “Estimation and Identification” in the main text.
3. We recover $\{\hat{\iota}\}$ along with the quarter fixed effects in Equation 3 as follows:
 - (a) Recover $\hat{\zeta}$ from estimates of Equation 2; i.e., $\hat{\zeta}_{kt} = \hat{\phi}_k^D \times \hat{\Delta}\zeta_{kt}$
 - (b) Construct the non-frontier group-level price index $\hat{P}_{Nt} = \left[\sum_{k \in \mathcal{N}_t} \left(\frac{\hat{P}_{kt}}{\hat{\zeta}_{kt}} \right)^{1-\hat{\eta}} \right]^{\frac{1}{1-\hat{\eta}}}$. Note that for the frontier technology $\hat{P}_{Ft} = \hat{P}_{kt}$ is obtained from the previous step
 - (c) Construct the non-frontier group-level quantity index $\hat{Q}_{Nt} = \left[\sum_{k \in \mathcal{N}_t} (\hat{\zeta}_{kt} \hat{Q}_{kt})^{\frac{\hat{\eta}-1}{\hat{\eta}}} \right]^{\frac{\hat{\eta}}{\hat{\eta}-1}}$. Note that for the frontier technology $\hat{Q}_{Ft} = \hat{Q}_{kt}$ is obtained from the previous step
 - (d) Estimate $\{\hat{\iota}, \hat{\rho}_t^D\}$ via Equation 3 using 2SLS. The instruments used are described under “Estimation and Identification” in the main text.
4. For the purpose of recovering marginal costs, we evaluate the continuation terms using realized future quantities and costs. We then solve Equation 6 backward, imposing a zero continuation value beyond the end of the sample, so that the terminal period first-order condition is static, i.e., $p_{jkT} = \frac{\tilde{\varepsilon}_{jkT}}{\tilde{\varepsilon}_{jkT}-1} \hat{c}_{jkT}$ at terminal value T . From this, backwards-induction generates the sequence of marginal costs $\{c_{jkt}\}_{t=1}^T$ using $\{q_{jkt}\}_{t=1}^T$, revenue shares entering Equation 8 in every period (s_{jkt} , s_{kt} , s_{jNt} , and s_{jt}), $\hat{\sigma}$, $\hat{\eta}$, $\hat{\iota}$, and Equation 6.

5. Estimate $\{\widehat{\nu}_j^S, \widehat{\phi}_k^S, \widehat{\rho}_t^S, \widehat{\beta}, \widehat{\kappa}\}$ and recover supply-side structural errors $\{\widehat{u}_{jkt}\}$ using a log-linear representation of Equation 4:

$$\log(\widehat{c}_{jkt}) - \gamma \log(H_{jkt}) = \beta \log(\bar{z}_{jkt}) + \kappa \log(w_{jt}) + \log(\nu_j^S) + \log(\phi_k^S) + \log(\rho_t^S) + u_{jkt}$$

6. Construct orthogonality moments, $E[\widehat{u}|Z^S] = 0$. The instruments for supply (Z^S) use the fact that firms make pricing decisions given their period t information set. At the optimal parameters, the structural errors are orthogonal to the information set. The supply-side instruments are described under “Estimation and Identification” in the main text.

A.3 Estimation Results

In this appendix we report select parameter estimates. Both the demand and supply-side specifications include a large set of fixed effects. The notes under each table indicate which fixed effects are included in each case. Table A.1 presents the estimation results for the demand side. Table A.2 presents GMM estimation results for three different specifications of learning-by-doing.²⁴

24. An implication of our model is that demand estimates are not affected by how we specify the learning process. This is because the buyers are assumed to be static utility maximizers.

Table A.1: Demand Estimation Results

PARAMETER	ESTIMATE	SE
LOWER NEST		
PRICE ($-\sigma$)	-16.358	(1.584)
QUALITY INDEX	4.315	(0.967)
N. OBS	1,847	
MIDDLE NEST		
TECHNOLOGY PRICE INDEX ($-\eta$)	-10.007	(1.904)
N. OBS	430	
UPPER NEST		
GROUP PRICE INDEX ($-\iota$)	-4.677	(0.286)
N. OBS	76	

Notes: The table presents 2SLS demand estimates and their standard errors based on the GSA Semiconductor Wafer Pricing Survey (2004-2019). The specification for the lower nest includes technology-time (technology-quarter) fixed effects, as well as country fixed effects. The specification for the middle nest includes technology fixed effects. The specification for the upper nest includes time (quarter) fixed effects. Prices in the lower nest are instrumented using wages, the experience of the firm in the same technology over the last three years, and the experience of the firm over the last three years in all technologies. The price index in the medium nest is instrumented using the average of the instruments used in the lower nest at the technology-quarter level. The price index in the upper nest is instrumented using the average of the instruments used in the lower nest across the suppliers in each group in a given quarter. Wages are quarterly average hourly manufacturing wages collected from each country's local statistical bureau and converted to US dollars using nominal exchange rates from [Feenstra, Inklaar, and Timmer \(2015\)](#). Prices and wages are in 1982 dollars using the consumer price index from FRED.

Table A.2: GMM Estimation Results

PARAMETER	(1)		(2)		(3)	
	ESTIMATE	SE	ESTIMATE	SE	ESTIMATE	SE
WAGES	0.020	(0.026)	-0.014	(0.023)	0.009	(0.023)
QUALITY INDEX	0.389	(0.057)	0.271	(0.064)	0.261	(0.067)
TECHNOLOGY LEARNING (γ)	-0.054	(0.006)	-0.181	(0.045)	-0.273	(0.066)
FIRM LEARNING (α)			0.190	(0.076)	0.137	(0.074)
WORLD LEARNING (μ)					0.121	(0.061)
N. OBS	1,847		1,847		1,847	
LEARNING RATE	3.7%		11.8%		17.3 %	
[95% C.I.]	[2.9%, 4.4%]		[6.2%, 17.0%]		[9.5%, 24.4%]	

Notes: The table presents the supply-side GMM estimates based on the GSA Semiconductor Wafer Pricing Survey (2004-2019). All specifications include time (quarter), technology, and country fixed effects. In specification (1), foundries learn only from past experience in the same technology. Specification (2) allows learning from experience in other technologies, and specification (3) – our baseline specification – additionally allows learning from world experience in the same technology. The instrument set in specification (1) includes lagged quantities sold and experience in the same technology. Specification (2) augments this set with experience in all technologies. Specification (3) further adds world experience in the same technology for technologies smaller than or equal to 180 nm (the reason for the focus on technologies 180 nm and below is that – as noted earlier – we exploit the order of entry, and in particular the lower prices of follower countries at entry, to identify the impact of cross-border learning on costs; however, given that our sample starts in the 2000s, we do not observe entry for technologies that are coarser than 180 nm, as such technologies are available by all suppliers in our sample); the number of quarters since technology introduction interacted with a follower-foundry indicator; lagged market share; the number of competitors in the previous quarter divided by the number of quarters the technology had been on the market; the distance to technologies previously produced by the supplier weighted by market share; and the exogenous regressors. Wages are quarterly average hourly manufacturing wages collected from each country’s local statistical bureau and converted to US dollars using nominal exchange rates from [Feenstra et al. \(2015\)](#). Wages are in 1982 dollars using the consumer price index from FRED. Standard errors are heteroskedasticity-robust GMM standard errors.

A.4 Robustness

The formulation of the supply side can accommodate several models of competition. In every case, equations (6)-(7) hold as stated—the dynamic wedge Λ_{jkt} and its recursion (7) are conduct-neutral, since an extra unit of output builds experience identically under any mode of competition—and only the mapping from the demand system into the perceived elasticity $\tilde{\varepsilon}_{jkt}$ changes. Specifically:

- **Quantity Competition:**

$$\tilde{\varepsilon}_{jkt} = \begin{cases} \left[\frac{1}{\sigma} + \left(\frac{1}{\eta} - \frac{1}{\sigma} \right) s_{jkt} + \left(\frac{1}{\iota} - \frac{1}{\eta} \right) S_{jNt} + \left(1 - \frac{1}{\iota} \right) S_{jt} \right]^{-1} & \text{if } k \in \mathcal{N}_t, \\ \left[\frac{1}{\sigma} + \left(\frac{1}{\iota} - \frac{1}{\sigma} \right) s_{jkt} + \left(1 - \frac{1}{\iota} \right) S_{jt} \right]^{-1} & \text{if } k \in \mathcal{F}_t. \end{cases}$$

- **Price Competition:** Under price competition, supplier j chooses the price vector of its portfolio, $\{p_{jkt}\}_{k \in K_j}$, holding rivals' prices fixed. The first-order condition for each price couples the margins of *all* of the supplier's products:

$$0 = q_{jkt} + \sum_{r \in K_j} (p_{jrt} - c_{jrt} - \Lambda_{jrt}) \frac{\partial q_{jrt}}{\partial p_{jkt}}, \quad k \in K_j.$$

Because each supplier operates one product per technology, the system is indexed by the supplier's technologies. Let $J_{jt} \equiv [\partial q_{jrt} / \partial p_{jkt}]_{r, k \in K_j}$ denote the supplier's demand Jacobian, whose elements follow from the demand system (all shares are revenue shares). For $k \in \mathcal{N}_t$, then:

$$\frac{\partial q_{jrt}}{\partial p_{jkt}} = \frac{q_{jrt}}{p_{jkt}} \times \begin{cases} -\sigma + (\sigma - \eta) s_{jkt} + (\eta - \iota) s_{jkt} s_{kt} + (\iota - 1) s_{jkt} s_{kt} S_{Nt} & \text{if } r = k, \\ (\eta - \iota) s_{jkt} s_{kt} + (\iota - 1) s_{jkt} s_{kt} S_{Nt} & \text{if } r \neq k, r \in \mathcal{N}_t, \\ (\iota - 1) s_{jkt} s_{kt} S_{Nt} & \text{if } r \in \mathcal{F}_t, \end{cases}$$

and for $k \in \mathcal{F}_t$ (where $s_{kt} = 1$ by construction),

$$\frac{\partial q_{jrt}}{\partial p_{jkt}} = \frac{q_{jrt}}{p_{jkt}} \times \begin{cases} -\sigma + (\sigma - \iota) s_{jkt} + (\iota - 1) s_{jkt} S_{Ft} & \text{if } r = k, \\ (\iota - 1) s_{jkt} S_{Ft} & \text{if } r \in \mathcal{N}_t. \end{cases}$$

Each element retains only the nest layers that products r and k share—the technology layer on the diagonal, the group layer if both are non-frontier, and the industry layer (the Cobb–Douglas closure) always. Stacking the first order conditions for each price over $k \in K_j$ and solving, yields:

$$p_{jkt} - c_{jkt} - \Lambda_{jkt} = \left[- (J'_{jt})^{-1} q_{jt} \right]_k,$$

where q_{jt} collects $(q_{jrt})_{r \in K_j}$ and $[\cdot]_k$ denotes the element corresponding to technology k . The perceived elasticity consistent with Equation 6 is therefore defined implicitly by:

$$\frac{1}{\tilde{\varepsilon}_{jkt}} \equiv \frac{\left[- (J'_{jt})^{-1} q_{jt} \right]_k}{p_{jkt}},$$

with which Equation 6–7 and the terminal condition apply without modification. Note that $\tilde{\varepsilon}_{jkt}$ is an *effective* elasticity—a scalar representation of the margin implied by the joint pricing system—and not the direct own-price elasticity. The two coincide only for a single-product supplier.

- **Monopolistic Competition:** Suppliers are atomistic, and take the technology-, group-, and industry-level price indices as given. Hence the perceived elasticity is $\tilde{\varepsilon}_{jkt} = \sigma$, and the markup is $\frac{\sigma}{\sigma-1}$.
- **Homogeneous Cournot** (homogeneous products within technology k , $\sigma \rightarrow \infty$):

$$\tilde{\varepsilon}_{jkt} = \begin{cases} \left[\frac{1}{\eta} s_{jkt} + \left(\frac{1}{\iota} - \frac{1}{\eta} \right) S_{jNt} + \left(1 - \frac{1}{\iota} \right) S_{jt} \right]^{-1} & \text{if } k \in \mathcal{N}_t, \\ \left[\frac{1}{\iota} s_{jkt} + \left(1 - \frac{1}{\iota} \right) S_{jt} \right]^{-1} & \text{if } k \in \mathcal{F}_t. \end{cases}$$

Our results regarding learning rates, and in particular cross-border learning, are robust to these alternative assumptions. Below we report results for the first three cases in Table A.3.

A.4.1 Results for alternative competition models

In column 1, we report the results for our baseline specification: quantity competition with differentiated products. In column 2, we report results for price competition, and in column 3, for monopolistic competition. We did not estimate the specification corresponding to homogeneous product quantity competition, as the assumption of homogeneity is clearly violated in our data.

Table A.3: GMM Estimation Results for Other Market Structures

PARAMETER	(1)		(2)		(3)	
	ESTIMATE	SE	ESTIMATE	SE	ESTIMATE	SE
WAGES	0.009	(0.023)	0.078	(0.022)	0.097	(0.022)
QUALITY INDEX	0.261	(0.067)	0.283	(0.066)	0.263	(0.066)
TECHNOLOGY LEARNING (γ)	-0.273	(0.066)	-0.250	(0.064)	-0.287	(0.064)
FIRM LEARNING (α)	0.137	(0.074)	0.103	(0.074)	0.133	(0.077)
WORLD LEARNING (μ)	0.121	(0.061)	0.136	(0.061)	0.177	(0.070)
N. OBS	1,847		1,847		1,847	
LEARNING RATE	17.3%		15.9%		18.1%	
[95% C.I.]	[9.5%, 24.4%]		[8.3%, 22.9%]		[10.6%, 24.9%]	

Notes: The table presents the supply-side GMM estimates based on the GSA Semiconductor Wafer Pricing Survey (2004-2019). All specifications include time (quarter), technology, and country fixed effects. In specification (1) we assume quantity competition with differentiated products. In specification (2) we assume price competition. In specification (3) we assume monopolistic competition. In all specifications, foundries learn from past experience in the same technology, from experience in other technologies, and from world experience in the same technology. In all specifications, the instrument set includes lagged quantities sold and experience in the same technology; experience in all technologies; world experience in the same technology for technologies smaller than or equal to 180 nm; the number of quarters since technology introduction interacted with a follower-foundry indicator; lagged market share; the number of competitors in the previous quarter divided by the number of quarters the technology had been on the market; the distance to technologies previously produced by the supplier weighted by market share; and the exogenous regressors. Wages are quarterly average hourly manufacturing wages collected from each country's local statistical bureau and converted to US dollars using nominal exchange rates from [Feenstra et al. \(2015\)](#). Wages are in 1982 dollars using the consumer price index from FRED. Standard errors are heteroskedasticity-robust GMM standard errors.

A.5 Additional Tables for the Counterfactual Exercises

Table A.4: Effects of Reducing China's Cross-Border Learning: 11–49 nm Technology

Country	$\mu_{\text{China}, 11/49} = 0$			$\mu_{\text{China}, 11/49} = \hat{\mu}/2$		
	Mean ΔP (%)	Median ΔP (%)	Aggregate ΔQ (%)	Mean ΔP (%)	Median ΔP (%)	Aggregate ΔQ (%)
China	13.3175	13.3721	-86.6810	5.7011	5.7264	-58.8040
Europe	0.0165	0.0016	0.0177	0.0133	0.0029	0.0115
Japan	0.0345	0.0400	1.7149	0.0238	0.0275	1.1033
Singapore	0.0123	0.0002	0.0616	0.0081	0.0008	0.0595
South Korea	-0.0028	-0.0041	0.0910	0.0104	0.0089	-0.0843
Taiwan	0.0756	0.0057	-0.1809	0.0516	0.0064	-0.1347
United States	0.0698	0.0742	0.9432	0.0491	0.0522	0.5936

Notes: The table reports equilibrium percentage changes relative to the benchmark. In the first counterfactual, China's cross-border learning parameter is set to zero for the 11–49 nm technology only. In the second counterfactual, it is set to one-half of its estimated value, $\hat{\mu}/2$. All other country-technology learning parameters remain at their benchmark values. Mean and median price changes are unweighted across product-quarter observations. Aggregate quantity changes are calculated using summed quantities within each country-technology cell.

Table A.5: Effects of Reducing China's Cross-Border Learning: 50–89 nm Technology

Country	$\mu_{\text{China}, 11/49} = 0$			$\mu_{\text{China}, 11/49} = \hat{\mu}/2$		
	Mean ΔP (%)	Median ΔP (%)	Aggregate ΔQ (%)	Mean ΔP (%)	Median ΔP (%)	Aggregate ΔQ (%)
China	-0.0170	0.0124	0.6313	-0.0137	0.0059	0.4457
Europe	-0.0048	-0.0031	-0.0026	-0.0070	-0.0060	0.0277
Japan	-0.0007	-0.0005	0.0258	-0.0030	-0.0029	0.0240
Singapore	-0.0025	-0.0021	0.0113	-0.0053	-0.0044	0.0318
South Korea	-0.0011	-0.0009	-0.0040	-0.0060	-0.0062	0.0480
Taiwan	-0.0025	-0.0039	0.0463	-0.0036	-0.0044	0.0484
United States	0.0043	-0.0030	-0.0039	0.0005	-0.0039	0.0388

Notes: The table reports equilibrium percentage changes in the 50–89 nm technology relative to the benchmark. The 50–89 nm technology is not directly subjected to a change in μ ; these therefore represent indirect equilibrium reallocation effects generated by reducing China's cross-border learning parameter for 11–49 nm. All other parameters remain at their benchmark values. Mean and median price changes are unweighted across product-quarter observations. Aggregate quantity changes are calculated using summed quantities within each country-technology cell.

Table A.6: Rest-of-World Effects of a 10% US Production Subsidy With and Without US Cross-Border Learning

Technology	Mean ΔP (%)		Aggregate ΔQ (%)	
	Subsidy only	Subsidy + $\mu_{US} = 0$	Subsidy only	Subsidy + $\mu_{US} = 0$
11/49	-1.37	1.15	-4.12	3.36
50/89	-1.60	1.42	-3.97	3.22
90	-2.34	1.96	1.31	-0.85
130/140	-1.85	1.66	-4.40	2.30
150/180	-1.56	1.35	-6.36	6.17
250/350	-1.05	0.93	-1.82	1.48
400/600	-2.23	1.78	-7.06	1.34

Notes: The table reports equilibrium percentage changes relative to the benchmark for all countries other than the United States from 2015Q1 onward. “Subsidy only” denotes a permanent 10% reduction in the US cost shifter beginning in 2015Q1, with the cross-border learning parameter μ_{US} remaining at its benchmark value. “Subsidy + $\mu_{US} = 0$ ” denotes the same subsidy with US cross-border learning set to zero for all technologies beginning in 2015Q1. Mean price changes are unweighted across product-quarter observations. Aggregate quantity changes are computed using summed quantities within each technology.

B Additional Information On Mechanisms Of Cross-Border Knowledge Transmission

B.1 Data and Measurement of Cross-Firm Collaborations

Co-authoring in semiconductor process engineering. Our co-authorship measure is built from the publication outlets in which semiconductor process engineers report their work. The most important of these is *SPIE*, the international professional society for optics and photonics. Given the central importance of photolithography to chip manufacturing, SPIE's technical conferences, and above all the annual Advanced Lithography meeting, have long served as the semiconductor manufacturing industry's principal technical forum (Kapoor and McGrath, 2014), and the papers presented there are published as the SPIE conference proceedings. We complement the SPIE proceedings with three of the field's main peer-reviewed engineering journals: the *IEEE Transactions on Semiconductor Manufacturing* (fabrication processes), SPIE's *Journal of Micro/Nanolithography*, and, for design automation, the *IEEE Transactions on Computer-Aided Design*.²⁵ These venues host academic and industrial research side by side. Previous work supports the representativeness of SPIE: Kapoor and McGrath (2014) confirm through industry interviews and directories that all firms pursuing lithography R&D participate in SPIE conferences, and show that leading participants' annual article counts closely track their R&D expenditures and, more loosely, their patenting.

We retrieve papers from OpenAlex, an open bibliographic database that indexes scholarly publications together with structured information on each author's affiliation. From the four venues, we collect every paper published between 2010 and 2025 that matches one of twelve frontier technology search terms fixed before the search was run (e.g., "EUV resist," "gate-all-around"; the full list appears in the notes to Table 4). Firm and country information comes from the affiliations printed on the papers: OpenAlex maps each affiliation to the employing institution (classified as a company, a university, or a research institute) and records the country of the establishment. We aggregate establishments of the same firm so that, for example, ASML's Dutch and U.S. establishments count as a single firm whose authors are located in different countries. The search yields 7,794 papers (retrieved July 2026), of which

25. We depart here from Kapoor and McGrath (2014), who study lithography and therefore use the SPIE proceedings alone. We tested a wider set of publication venues, including the industry's two other major technical conferences (the International Electron Devices Meeting and the Symposium on VLSI Technology), but their papers and author affiliations are too incompletely indexed in our data source, so they were excluded. Ultimately, expanding our sample beyond SPIE does not add many more papers: SPIE accounts for 87% of the papers we examine below.

2,287 have at least one company-affiliated author; the latter form the sample on which our measures are defined.

Mentions of R&D collaborations in earnings calls. We develop a natural language processing (NLP) pipeline to identify instances of inter-firm R&D collaboration in the text of quarterly earnings call transcripts, obtained from the S&P Capital IQ module of Wharton Research Data Services (WRDS). The sample covers calls held between 2010 and 2025 by publicly listed firms in our sample. Transcripts are available as speaker turns: each record (or component) is a contiguous block of speech attributed to a named speaker, tagged by a unique transcript component ID, which can then be linked to various transcript, company, and speaker metadata. This includes the speaker’s name, role, and company, and whether the component belongs to the prepared remarks or the question-and-answer portion of the call. Our unit of observation is the `{transcriptid, transcriptcomponentid, componenttext}` tuple filtered by the latest `mostimportantdateutc` variable when there are multiple copies of the same transcript in the database.

Firms do not describe R&D collaborations in a standardized way, and the disclosures are scattered through calls whose primary content is financial. We therefore begin from an explicit definition, which serves both as the construct we measure and as the coding guideline the pipeline implements. A passage qualifies as an instance of R&D collaboration only if it (i) names a partner firm; (ii) uses language describing joint work; (iii) identifies a specific technical problem or objective the firms are working on together; and (iv) concerns a semiconductor-relevant technology. The definition is deliberately strict: it excludes collaborations mentioned in general terms, joint work whose counterparty is unnamed, and arm’s-length buyer-supply relationships.

Implementation proceeds in two stages. The first stage is a high-recall filter whose purpose is to reduce a corpus of $N = 84,438$ components to a tractable set of candidate passages, and which is intended to be permissive: we apply named entity recognition, specifically spaCy 3.8.7’s transformer-based `en_core_web_trf` to extract organization mentions. Detected organization names are first matched to our set of firms, building a dictionary of canonical names and known aliases per firm from the detections. This dictionary is then used to link to WRDS `companyname`, using RapidFuzzstring matching. After eliminating self-mentions, we retain the components in which a named firm co-occurs with a term from a hand-curated

lexicon of collaboration vocabulary.²⁶ This stage enforces criteria (i) and (ii) mechanically and leaves $N = 2,904$ candidate sentences from $N = 1,292$ transcript components.

The second stage adjudicates the remaining criteria through three prompted passes to Claude Sonnet 4.6 (`claude-sonnet-4-6`) with the temperature set to 0. The model’s training data cutoff is January 2026, which postdates our sample period. All requests were issued on August 25th, 2026. The first pass assesses whether two or more firms are described as jointly working on a shared R&D problem. The remaining passes enforce internal validity checks ensuring that the different components of our definition all refer to the same part of text (e.g., the named firm is the one collaborating on the R&D problem identified at a different stage of the pipeline). Table B.1 contains positive and negative examples from each pass.

The resulting measure is a lower bound on collaboration: it counts collaborations that were disclosed, disclosed with the partner named, and disclosed with technical specificity. This is worth bearing in mind when we compare the network of collaboration obtained in this way from earnings calls to those derived from co-authorship patterns in the engineering literature.

Co-patenting. We measure co-patenting rates in our sample for the years 2010-2025 using data from PatentsView, the United States Patent and Trademark Office’s research database of granted US patents. We record two firms as co-patenting when both appear as separate assignees on the same granted patent.²⁷ The share is the sum of co-patents between firms in the row and column category, normalized by the total granted patents held by firms in the row category.

B.2 Data and Measurement of Movement of Engineers Within and Across Firms

Fab technology and upgrade events. We identify technology upgrades from SEMI’s World Fab Watch a proprietary census of front-end semiconductor manufacturing facilities. The unit of observation is the fab, recorded quarterly, with information on the facility’s minimum feature size (the technology node), installed capacity, and product type. A fab is recorded as upgrading in the quarter in which its feature size drops. We aggregate to the firm–country–

26. We use spaCy’s `PhraseMatcher` with the `en_core_web_sm` model to identify sentences containing any expression from a predefined dictionary of inter-firm relationship terms and phrases, matching on lemmatized forms to capture morphological variants. The predefined dictionary consists of the following seed terms: partnership, strategic partnership, design partner, development partner, contractual, collaboration, working with, supporting, collaborate, strategic, collaborative effort, collaborative, relationship, engagement, cooperation, partner, co operation, co-operation, jointly develop, alliance, joint venture, and joint-venture.

27. We consider assignees, the entity that owns the patent’s rights, rather than inventor or applicant which are typically individuals or legal firms.

Table B.1: Examples from the three-pass classification pipeline

Retained	Dropped
<i>Pass 1 — collaboration identification.</i> Is the paragraph describing a specific joint undertaking between named firms, rather than a customer relationship or a general characterization of a business tie?	
3647917__01 — Cadence “... we are collaborating with Xilinx, ARM and TSMC to build the industry first test chip for Cache Coherent Interconnect for Accelerators or CCIX, incorporating Cadence IP and using Cadence tools on the TSMC 7-nanometer FinFET process.” A named deliverable and named parties, each contributing something distinct.	3670510__01 — United Microelectronics “... Allegro MicroSystems, a leader in high-performance power and sensor solutions, signed a multi-year manufacturing agreement with UMC to ensure long-term capacity support for their growing wafer requirement in industrial and automotive segment.” This describes a supplier relationship with no distinct technical contribution to a shared problem.
<i>Pass 2 — production relevance.</i> Does the identified collaboration concern semiconductor production, design, R&D, or supply-chain operations, and is the speaking firm one of its parties?	
3545923__01 — Cadence “We collaborated with Imec, an international research and innovation hub, on the industry-first 3-nanometer cache chip tapeout using Cadence Innovus Implementation Systems and Genus Synthesis Solution.” Design-enablement work with a research institute at an advanced node, with Cadence named as one of the two parties.	159395__01 — Nvidia “NVIDIA and Intel announced a collaboration to bring high-performance 3D gaming and multimedia platform to handheld devices ... combines the NVIDIA GoForce family of handheld GPUs with Intel’s newest processor family, codenamed Monahans.” Unrelated to innovating in the production of semiconductors.
	2407809__02 — ASML “... Toshiba and Canon announced that they would be collaborating on nano and print technology on production for 1z-nanometer node.” A production collaboration, but between two other firms.
<i>Pass 3 — consistency check.</i> Do pass 1’s problem statement and pass 2’s explanation describe the same collaboration?	
5136148__01 — United Microelectronics <i>Pass 1:</i> production of a power semiconductor at USJC’s 300 millimeter fab to serve growing demand in the automotive market. <i>Pass 2:</i> USJC, UMC’s Japan subsidiary, and DENSO collaborating on power semiconductor production at USJC’s 300 mm fab. The two texts refer to the same instance of collaboration.	5985007__01 — Amkor <i>Pass 1:</i> offering a seamless manufacturing footprint in Europe to support the European supply chain. <i>Pass 2:</i> a wafer bump and wafer probe arrangement with GlobalFoundries at its Drayson factories. Different. Pass 1 recorded a broad European footprint initiative while pass 2 identified a specific services arrangement with a named partner.
<i>Notes:</i> Identifiers are <code>collab_id</code> , formed as the paragraph identifier and a collaboration number. Quotations for retained records and for records dropped at pass 2 are the verbatim evidence field returned by pass 1; records rejected at pass 1 return no evidence field, so that excerpt is drawn directly from the transcript. Pass 3 entries paraphrase the <code>collaboration_problem</code> and <code>production_explanation</code> fields. Two of the 373 records reaching pass 3 were labelled <i>different</i> ; no record was labelled <i>unclear</i> .	

year: a firm–country is treated in the year in which the smallest feature size across its fabs in that country falls relative to the previous year. Figure B.1 plots the sample of firm-country upgrading events. The figure shows how the global technology frontier advances during this period, and it also plots the many firm-country pairs experiencing technology upgrading events within this technology frontier. These upgrading events will be central to our research design.

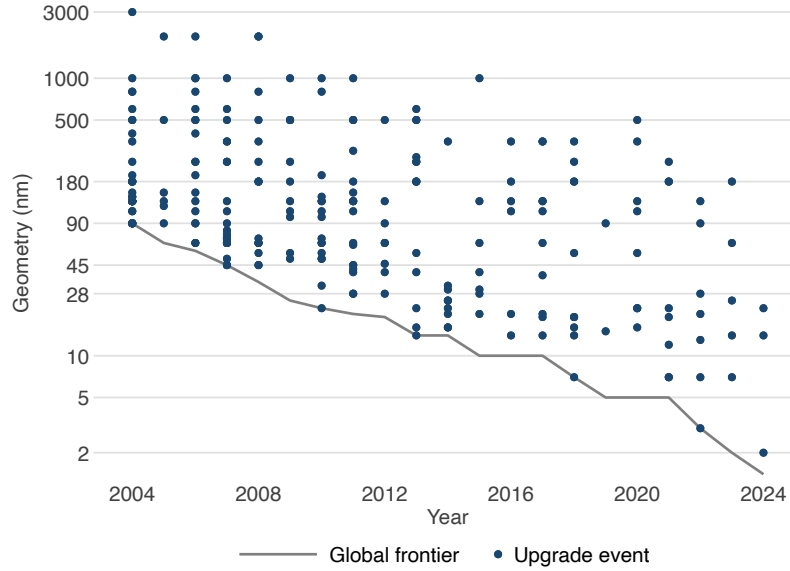


Figure B.1: Chips Technology Frontier and Firm Upgrading events

Cross-border movement of engineers. We measure labor flows using Revelio Labs’ individual positions data, accessed through Wharton Research Data Services. Revelio assembles standardized employment histories from public professional profiles; each record is a position held by one individual, with the employer, the country and city of the establishment, start and end dates, and two model-assigned attributes we use in our analysis: an occupational role drawn from a taxonomy of roughly 1,500 clusters, and a seniority score on a one-to-seven scale.²⁸ We retrieve every position at every firm Revelio classifies to NAICS 334413: “semiconductor and related device manufacturing,” resulting in 6,092 firms and 3.77 million position records.

We define an arrival as *cross-border* if the previous reported position was located in a different country.²⁹ As in Section B.1, a worker’s country is the location of the establishment

28. Revelio predicts seniority from the job title, company, industry, age, previous seniority and position history; one is the most junior and seven the most senior. Role clusters are likewise assigned from profile text. Both are estimates, not administrative records.

29. Our Revelio data extract only contains positions within NAICS 334413, raising the possibility that we misclassify some cross-border moves. In robustness checks, we have experimented with a different

at which the position is held, not the headquarters of the employer: an engineer moving from Intel’s Oregon site to Intel’s Israeli site is a cross-border move, while a move between two firms’ Californian sites is not.

To capture workers whose human capital may contain the process engineering knowledge useful to an upgrading fab, we construct four measures of cross-border worker flows, ordered from the most agnostic to the most restrictive. Throughout we retain only workers at seniority three or above, which removes entry-level and intern positions.

The first measure retains any arrival whose occupation string contains “engineer,” and we refer to these workers as *Senior Engineers*. The second adds an origin restriction, keeping only those senior engineers whose previous employer appears in the SEMI fab data described above, so that the worker is known to have moved from a firm that operates fabs; we refer to these as *Senior Engineers SEMI*.

The remaining two measures add occupational content, using the role clusters the taxonomy assigns while leaving in place the restriction that the worker arrives from a firm listed in the SEMI data. A relatively broad taxonomy retains sixteen clusters tied directly to fab operations — process, manufacturing and production, and equipment engineering — together with twenty-seven adjacent clusters covering test, quality and reliability, materials and chemistry, automation, equipment field applications, and research, device and product engineering. We refer to these as *Senior Core and Related SEMI*. The fourth drops the adjacent clusters and keeps only the sixteen tied to fab operations; it is the most restrictive measure we use, and the one for which the mapping from occupation to process knowledge is most direct. This measure is labeled *Senior Core SEMI*.³⁰

The estimation sample. We build a crosswalk between firm–country pairs in SEMI and in Revelio, yielding a firm–country–year panel of cross-border labor inflows and upgrading events over 2004–2024.³¹ The estimation sample contains 236 upgrading events. Summary statistics are reported in Table B.2.

definition of cross-border moves based on whether a worker held a position abroad within 2 years of their current position. The results are stable.

30. Both of these last two also restore nine generic labels — “engineer,” “manufacturing” and the like — which the taxonomy assigns to profiles too sparse to classify further, and which would otherwise discard fab engineers whose recorded titles are uninformative.

31. The panel is annual because many Revelio positions report no start date finer than the year.

Table B.2: Summary statistics: cross-border senior labor flows in the event-study samples

	Mean	N	SD	p25	p50	p75	# events	# clusters
<i>Panel A: Technology upgrading events</i>								
Senior Engineer	2.72	3068	9.27	0.00	0.00	2.00	236	20258
Senior Engineers SEMI	2.19	3068	7.97	0.00	0.00	1.00	236	20258
Senior Core and Related SEMI	1.19	3068	3.87	0.00	0.00	1.00	236	20258
Senior Core SEMI	0.65	3068	2.33	0.00	0.00	0.00	236	20258
<i>Panel B: Expansion placebo</i>								
Senior Engineer	1.52	2543	4.32	0.00	0.00	1.00	126	10539
Senior Engineers SEMI	1.15	2543	3.52	0.00	0.00	1.00	126	10539
Senior Core and Related SEMI	0.63	2543	2.03	0.00	0.00	0.00	126	10539
Senior Core SEMI	0.34	2543	1.28	0.00	0.00	0.00	126	10539
<i>Panel C: Within vs. across firms</i>								
All cross-country	2.72	3068	9.27	0.00	0.00	2.00	236	20258
Within firm	1.74	3068	6.83	0.00	0.00	1.00	236	20258
Across firms	0.98	3068	3.03	0.00	0.00	1.00	236	20258

Notes: Summary statistics for the outcome variables shown in the event-study figures, computed on the corresponding estimation samples. Each sample is a stacked event-study panel following [Cengiz et al. \(2019\)](#): each stack is centered on a treated firm-country with an event window of $[-3, +2]$ years, and clean controls are firm-countries experiencing no event within the window. In Panels A and C the event is a firm-country improving its smallest production geometry (technology upgrade); in Panel B the event is a new fab opening at the firm-country with no geometry upgrade anywhere in the event window (expansion placebo). TSMC Taiwan is excluded throughout: event windows where it is the treated unit are dropped, and it never qualifies as a clean control. Outcome variables count senior workers arriving at a firm-country from abroad (immediately prior position in a different country), measured at the firm-country $\times$ year level, under successively different definitions: workers whose role title contains “engineer” (*Senior Engineer*); engineers whose prior position was at a firm appearing in the SEMI fab-level production database, matched directly or through its ultimate parent (*Senior Engineers SEMI*); workers arriving from a SEMI-database firm in one of 52 role titles spanning fab process operations and adjacent technical functions (*Senior Core and Related SEMI*); and the subset of the preceding measure in the 25 role titles most directly tied to fab process operations (*Senior Core SEMI*). Panel C decomposes cross-country moves of senior engineers (all origins) into moves between establishments of the same firm (*Within firm*) and moves across firms (*Across firms*). Clean controls may appear in multiple stacks, the stacked panel repeats firm-country $\times$ year observations; Mean, SD, and percentiles are therefore computed on the estimation sample deduplicated to one observation per firm-country $\times$ year, and N reports the number of unique firm-country $\times$ year cells. # events is the number of distinct event windows (stacks) and # clusters the number of distinct firm-country $\times$ event clusters (the level of standard-error clustering in the regressions), both counted on the full stacked estimation sample before deduplication; they are identical across rows within a panel because all outcomes share the same sample. Data sources: SEMI Transactions Fab-Level Production Data; Revelio Labs Individual Positions Data.

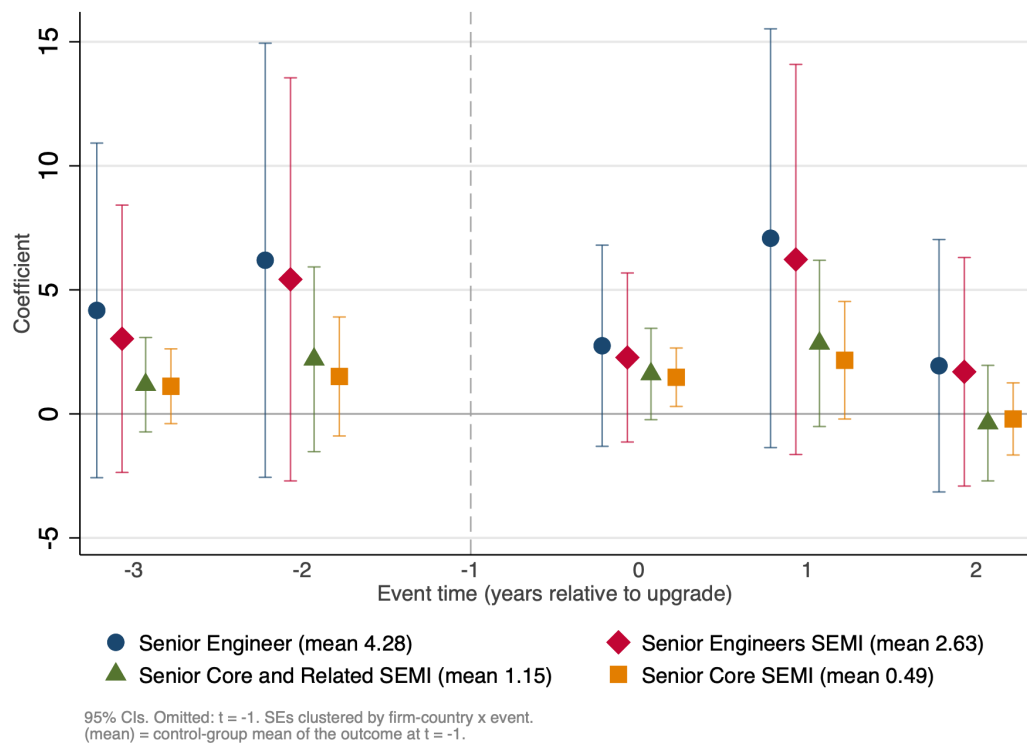


Figure B.2: Cross-border skilled, senior labor flows around technology upgrading events; Logic Fabs only

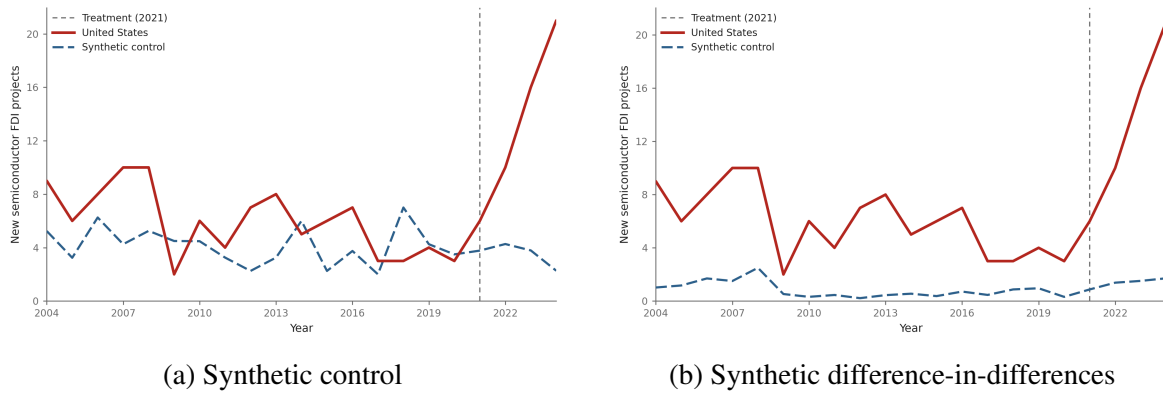


Figure B.3: US semiconductor FDI vs. synthetic counterfactual

Notes: The outcome is the annual count of announced greenfield FDI projects by destination country, from fDi Markets, restricted to cross-border projects in the semiconductor machinery and semiconductors & other electronic components sub-sectors. The panel covers 2004–2024, with the United States as the treated unit and 2021 as the first treated year. Panel (a) reports the synthetic control estimator of [Abadie, Diamond, and Hainmueller \(2010\)](#), which places weight on two donors, the United Kingdom (0.75) and Germany (0.25), with an ATT of 9.73 additional projects per year (SE 2.26). Panel (b) reports the synthetic difference-in-differences estimator of [Arkhangelsky, Athey, Hirshberg, Imbens, and Wager \(2021\)](#), which spreads weight across eleven donors, the largest being Switzerland (0.20), France (0.15) and Austria (0.13), with an ATT of 5.96 (SE 2.01). For both panels, standard errors are from placebo inference, following [Arkhangelsky et al. \(2021\)](#): the treated unit is discarded, each of the 19 donors is in turn reassigned as treated at the same date, and the estimator is re-run; the standard error is the standard deviation of the 19 resulting placebo estimates.

References

- Abadie, A., A. Diamond, and J. Hainmueller (2010). Synthetic control methods for comparative case studies: Estimating the effect of california’s tobacco control program. *Journal of the American statistical Association* 105(490), 493–505.
- Arkhangelsky, D., S. Athey, D. A. Hirshberg, G. W. Imbens, and S. Wager (2021). Synthetic difference-in-differences. *American Economic Review* 111(12), 4088–4118.
- Cengiz, D., A. Dube, A. Lindner, and B. Zipperer (2019). The effect of minimum wages on low-wage jobs. *The quarterly journal of economics* 134(3), 1405–1454.
- Feenstra, R. C., R. Inklaar, and M. P. Timmer (2015). The next generation of the penn world table. <https://www.rug.nl/ggdc/productivity/pwt/>. Version 10.01, retrieved from <https://www.rug.nl/ggdc/productivity/pwt/>.
- Kapoor, R. and P. J. McGrath (2014). Unmasking the interplay between technology evolution and R&D collaboration: Evidence from the global semiconductor manufacturing industry, 1990–2010. *Research Policy* 43, 555–569.